King Fahad Central Hospital

STROKE UNIT

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POLICY AND PROCEDURE MANUAL

Aligned with American Heart Association / American Stroke Association (AHA/ASA)

Scientific Guidelines and American Stroke Association International Comprehensive Stroke Center Certification Standards

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Approved By: Medical Director / Stroke Program Medical Director

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Manual Introduction and Scope

This manual constitutes the official Policy and Procedure framework for the Stroke Unit and Stroke Program of King Fahad Central Hospital. It is developed to ensure consistent, evidence-based, multidisciplinary stroke care across the continuum — pre-hospital notification, emergency department triage, acute treatment, inpatient management, rehabilitation, and secondary prevention — in alignment with the scientific guidelines of the American Heart Association / American Stroke Association (AHA/ASA) and the certification standards of the American Stroke Association International Comprehensive Stroke Center Certification program.

Each policy in this manual follows a standardized format — Purpose, Definitions, Policy, Procedure, References, and Attachments — with a policy identification header (policy number, effective date, review date, approving authority, and applicability) and a footer identifying the applicable accreditation standard and page numbering, to support survey readiness and document control.

This manual is reviewed and approved by the Stroke Committee and Medical Director at minimum annually, and immediately upon any material change in evidence-based guidelines, regulatory requirements, or certification standards. All bracketed fields (e.g., [Effective Date], [Review Date]) are to be completed by the organization prior to implementation.

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POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-001	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Program Medical Director		
APPLIES TO	Hospital Wide — Emergency Department, Stroke Unit, ICU, Radiology, Laboratory, Pharmacy, Rehabilitation		

SU-001 Stroke Program Organization, Governance and Staffing

1. Purpose

To define the organizational structure, governance, staffing, and multidisciplinary team requirements of the Stroke Program, and to establish the administrative framework required to support consistent, evidence-based stroke care in compliance with American Heart Association/American Stroke Association (AHA/ASA) guidelines and American Stroke Association International Comprehensive Stroke Center certification standards.

2. Scope

This policy applies to all departments and personnel involved in the identification, triage, treatment, admission, and rehabilitation of stroke and transient ischemic attack (TIA) patients, including the Emergency Department (ED), Stroke Unit, Intensive Care Unit (ICU), Neurology, Neurosurgery, Interventional Radiology, Diagnostic Imaging, Laboratory, Pharmacy, Nursing, and Rehabilitation Services (Physical Therapy, Occupational Therapy, Speech-Language Pathology).

3. Definitions and Abbreviations

3.1 AHA/ASA: American Heart Association / American Stroke Association.

3.2 ASA ISC: American Stroke Association International Stroke Center Certification.

3.3 PSC: Primary Stroke Center.

3.4 CSC: Comprehensive Stroke Center.

3.5 GWTG-Stroke: Get with the Guidelines®–Stroke, the AHA/ASA hospital-based quality improvement registry and program.

3.6 Stroke Medical Director: A physician (neurologist or physician with equivalent stroke expertise) responsible for the clinical oversight of the Stroke Program.

3.7 Stroke Coordinator: A registered nurse or allied health professional responsible for day-to-day program coordination, data abstraction, staff education, and performance improvement activities.

3.8 Multidisciplinary Stroke Team: The group of clinical disciplines required for comprehensive stroke care (see Section 5).

3.9 Interprofessional Committee (IPC): the multidisciplinary governance body — synonymous with the Stroke Committee referenced throughout this manual — responsible for overseeing the Stroke Program's mission, goals, scope, and organizational structure.

3.10 Telehealth Champion: A clinician with telehealth experience/expertise who identifies telehealth opportunities and drives adoption of telestroke and related telehealth services for the Stroke Program.

3.11 ASRC: Acute Stroke Ready Center — a facility providing initial stroke identification, stabilization, and, where indicated, transfer to a higher level of care.

4. Policy

4.1 The organization maintains a designated Stroke Program with defined governance, medical leadership, and administrative support, consistent with American Stroke Association International Comprehensive Stroke Center certification requirements and AHA/ASA scientific guidelines.

4.2 The Stroke Program operates under a Stroke Committee that meets at minimum quarterly to review performance data, complications, protocol deviations, and opportunities for improvement.

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SU-001	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Program Medical Director		
APPLIES TO	Hospital Wide — Emergency Department, Stroke Unit, ICU, Radiology, Laboratory, Pharmacy, Rehabilitation		

4.3 A Stroke Medical Director and a Stroke Program Coordinator are formally appointed in writing, with defined roles, responsibilities, and reporting lines to hospital administration and the Medical Staff.

4.4 All staff involved in stroke care receive stroke-specific orientation and ongoing education, with documented annual competency verification.

4.5 The organization participates in a standardized stroke data registry (e.g., GWTG-Stroke) for continuous quality measurement and public reporting where applicable.

4.6 The Stroke Unit maintains defined nurse-to-patient staffing ratios appropriate to patient acuity, with dedicated stroke-trained nursing staff on all shifts.

4.7 The Stroke Committee (IPC) develops and maintains a Stroke Program Charter documenting committee membership and roles, meeting frequency and attendance expectations, program goals, and facility demographics, reviewed and updated at minimum annually.

4.8 Program requirements are scaled to the certification level held or pursued by the organization (Acute Stroke Ready Center, Primary Stroke Center, or Comprehensive Stroke Center); where a requirement differs by certification level, this is noted within the applicable policy section.

4.9 A designated Telehealth Champion identifies and supports telestroke and related telehealth service opportunities in coordination with the Stroke Medical Director and Stroke Program Coordinator.

5. Multidisciplinary Stroke Team

The Stroke Team includes, at minimum, the following disciplines, each with defined roles documented in their respective privileging/competency files:

5.1 Stroke Medical Director (Neurologist).

5.2 Emergency Medicine Physicians.

5.3 Neurointerventionalist / Interventional Radiologist (for mechanical thrombectomy-capable and comprehensive centers).

5.4 Neurosurgery (on-call availability for hemorrhagic stroke and decompressive procedures).

5.5 Stroke Program Coordinator / Data Abstractor.

5.6 Stroke Unit Registered Nurses (NIHSS-certified).

5.7 Radiology / CT and MRI Technologists (24/7 availability).

5.8 Pharmacy (stroke order set review, thrombolytic preparation support).

5.9 Laboratory Services (STAT turnaround for stroke panel).

5.10 Physiatrist (Physical Medicine and Rehabilitation).

5.11 Physical Therapy, Occupational Therapy, and Speech-Language Pathology.

5.12 Case Management / Social Work.

5.13 Dietitian/Clinical Nutrition.

6. Staffing and Competency Requirements

6.1 All physicians and registered nurses providing direct stroke care must complete NIHSS certification (or equivalent) upon hire and recertify per organizational policy (minimum every 2 years or per certifying body requirement).

6.2 Stroke Unit nursing staff must complete a stroke-specific orientation program covering the acute stroke pathway, thrombolytic administration and monitoring, dysphagia screening, and post-stroke complication recognition prior to independent practice.

6.3 Physicians providing thrombolytic therapy must hold documented privileges for IV alteplase/tenecteplase administration.

6.4 A minimum staffing complement and on-call schedule is maintained to support 24/7/365 stroke code response, CT imaging, laboratory turnaround, and (where applicable) endovascular intervention.

7. Committee Structure and Performance Review

7.1 The Stroke Committee reviews, at minimum quarterly: door-to-CT times, door-to-needle times, door-to-groin puncture times (thrombectomy-capable/comprehensive centers), thrombolytic complication rates, GWTG-Stroke measure compliance, and mortality/morbidity outliers.

7.2 Performance data are benchmarked against AHA/ASA Target: Stroke® goals and applicable American Stroke Association International Comprehensive Stroke Center certification standards.

7.3 Identified gaps trigger a documented performance improvement action plan with assigned ownership and follow-up review.

8. References

8.1 American Heart Association/American Stroke Association. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke.

8.2 American Stroke Association. International Comprehensive Stroke Center Certification Standards.

8.3 AHA/ASA Get With The Guidelines®–Stroke Program Manual.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-002	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director		
APPLIES TO	Emergency Department, Stroke Unit, ICU, Inpatient Wards, Radiology		

SU-002 Stroke Code Activation — Pre-Hospital and In-Hospital

1. Purpose

To standardize the rapid identification, notification, and response process ("Stroke Code") for any patient with suspected acute stroke, in order to minimize time from symptom onset/last known well to treatment, consistent with AHA/ASA Target: Stroke® time goals.

2. Definitions and Abbreviations

2.1 Last Known Well (LKW): The time the patient was last observed to be at their baseline neurological function.

2.2 FAST: Face, Arms, Speech, Time — a rapid stroke screening mnemonic.

2.3 BE-FAST: Balance, Eyes, Face, Arms, Speech, Time — an expanded screening tool incorporating posterior circulation signs.

2.4 Stroke Code / Code Stroke: The formal activation process that mobilizes the multidisciplinary stroke response team.

2.5 NIHSS: National Institutes of Health Stroke Scale.

2.6 EMS: Emergency Medical Services.

3. Policy

3.1 A Stroke Code may be activated by any nurse, physician, or EMS provider identifying signs/symptoms consistent with acute stroke using the BE-FAST screening tool, regardless of LKW time, unless clearly outside the therapeutic window for all interventions and the patient is not a candidate for any acute therapy.

3.2 Pre-hospital EMS providers use a validated stroke screening tool and provide advance hospital notification ("pre-notification") for all suspected stroke patients, including LKW time, screening result, vital signs, and estimated time of arrival.

3.3 Pre-notified patients are met by the Stroke Team at the point of arrival and transferred directly to the CT scanner or a designated resuscitation/stroke bay.

3.4 Time targets (AHA/ASA Target: Stroke® Phase III goals, used as internal benchmarks):

3.4.1 Door-to-physician evaluation: within 10 minutes of arrival.

3.4.2 Door-to-CT initiation: within 20 minutes of arrival (AHA/ASA and institutional benchmark).

3.4.3 Door-to-CT interpretation: within 45 minutes of arrival.

3.4.4 Door-to-needle (IV thrombolytic administration): within 60 minutes of arrival for ≥85% of patients (goal ≤45 minutes for ≥50% of patients).

3.4.5 Door-to-groin puncture (mechanical thrombectomy): within 90 minutes of arrival (direct presentation) or 60 minutes (transfer patients), per institutional thrombectomy-capable/comprehensive stroke center goals.

3.4.6 EMS-to-stroke-unit or ICU admission decision within 3 hours of arrival.

4. Procedure — In-Hospital Stroke Code Activation

4.1 Any staff member identifying a patient with acute focal neurological deficit(s) and LKW within the treatment window activates the Stroke Code by calling the designated hospital stroke code number/overhead page.

4.2 The triage nurse or first responder immediately:

4.2.1 Notes and documents the exact LKW time (not the time the deficit was discovered).

4.2.2 Performs BE-FAST screening.

4.2.3 Obtains vital signs including finger-stick glucose.

4.2.4 Transfers the patient to a resuscitation/critical care bay.

4.2.5 Obtains IV access (18-gauge preferred) and draws STAT labs (CBC, basic metabolic panel/renal profile, coagulation studies [PT/INR, aPTT], troponin, and type & crossmatch/type & screen as per protocol).

4.3 The Stroke Code activation immediately and simultaneously notifies:

4.3.1 Emergency Physician / Stroke Team physician on duty.

4.3.2 On-call Neurologist / Stroke Medical Director.

4.3.3 CT Technologist and Radiology (to prioritize immediate imaging).

4.3.4 Stroke Unit / ICU charge nurse (bed availability).

4.3.5 Pharmacy (thrombolytic preparation on standby).

4.3.6 Laboratory (STAT stroke panel priority).

4.4 The responding physician performs a focused history and examination, including NIHSS scoring, and reviews the IV thrombolytic inclusion/exclusion checklist.

4.5 The patient is transported directly to CT for non-contrast head CT ± CT angiography (arch to vertex) ± CT perfusion, per the Acute Ischemic Stroke Management policy (SU-003).

4.6 All Stroke Code time points (activation time, physician arrival, CT arrival, CT completion, CT read, needle time, groin puncture time as applicable) are documented on the Stroke Code / Acute Stroke Flow Sheet for subsequent data abstraction and performance review.

4.7 In-hospital (“wake-up” or inpatient-onset) strokes follow the identical activation and time-target process; the LKW is the last time the inpatient was observed at baseline.

5. Procedure — Pre-Hospital Notification

5.1 EMS pre-notification calls are routed directly to the Emergency Department charge nurse/Stroke Team.

5.2 The receiving nurse relays LKW time, screening findings, vital signs, blood glucose (if obtained), anticoagulant use, and estimated arrival time to the Stroke Team.

5.3 A resuscitation bay and CT scanner are reserved in advance of patient arrival whenever feasible.

6. Documentation

All Stroke Code events are documented in the Nursing Acute Stroke Screening Form and Emergency Response Team Initial Assessment Form (see Attachments), including LKW time, activation time, and each subsequent time-stamped milestone.

7. References

7.1 AHA/ASA Target: Stroke® Phase III Initiative.

7.2 AHA/ASA 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke.

7.3 American Stroke Association International Stroke Center Certification quality and performance requirements.

8. Attachments

8.1 Stroke Code / Acute Stroke Flow Sheet.

8.2 Nursing Acute Stroke Screening Form (BE-FAST).

8.3 EMS Pre-Notification Communication Form.

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SU-003	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Medical Director		
APPLIES TO	Hospital Wide		

SU-003 Acute Ischemic Stroke Management — IV Thrombolysis and Mechanical Thrombectomy

1. Purpose

To delineate the evidence-based steps for the assessment, imaging, and treatment (intravenous thrombolysis and/or mechanical thrombectomy) of patients presenting with acute ischemic stroke, aligned with the 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke, which supersedes the 2018/2019 guideline previously referenced in this policy.

2. Definitions and Abbreviations

2.1 IV tPA / Alteplase: Recombinant tissue plasminogen activator, an IV thrombolytic agent.

2.2 Tenecteplase (TNK): An IV thrombolytic agent administered as a single bolus; for adult acute ischemic stroke, when selected under the approved institutional protocol, the dose is 0.25 mg/kg IV (maximum 25 mg).

2.3 NIHSS: National Institutes of Health Stroke Scale.

2.4 LKW: Last Known Well.

2.5 ASPECTS: Alberta Stroke Program Early CT Score, used to quantify early ischemic changes on non-contrast CT.

2.6 CTA: CT Angiography. CTP: CT Perfusion.

2.7 MT: Mechanical Thrombectomy (endovascular therapy).

2.8 LVO: Large Vessel Occlusion.

3. Policy

3.1 IV thrombolytic therapy improves functional outcome when administered within the appropriate therapeutic window and is the standard of care for eligible patients with disabling acute ischemic stroke.

3.2 Door-to-needle time is a primary quality metric and should be ≤60 minutes for ≥85% of treated patients (institutional goal ≤45 minutes where feasible), consistent with AHA/ASA Target: Stroke®.

3.3 Mechanical thrombectomy is offered to eligible patients with confirmed large vessel occlusion, in coordination with a certified neurointerventional service (on-site or via a defined transfer agreement with a Comprehensive Stroke Center / Thrombectomy-Capable Stroke Center).

3.4 Eligibility for both IV thrombolysis and mechanical thrombectomy is determined using current AHA/ASA inclusion/exclusion criteria, applied by a physician credentialed in acute stroke management.

3.5 Treatment decisions and risks/benefits are discussed with the patient or legally authorized representative and documented as informed consent, consistent with organizational policy for emergency/time-critical procedures.

4. Procedure — Initial Assessment (0–6 Hours from Onset/LKW)

4.1 Following Stroke Code activation (SU-002), the responding physician performs a focused history, NIHSS examination, and reviews the IV thrombolytic inclusion/exclusion checklist.

4.2 Airway, breathing, and circulation are stabilized; supplemental oxygen is provided to maintain SpO_2 ≥ 94%.

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APPROVED BY	Medical Director / Stroke Medical Director		
APPLIES TO	Hospital Wide		

4.3 STAT labs are sent (CBC, renal profile, coagulation studies, glucose, troponin, type & screen); treatment is not delayed pending coagulation results unless the patient is on anticoagulants, has a bleeding disorder, or thrombocytopenia is suspected.

4.4 Non-contrast head CT is obtained emergently (goal: CT initiated within 20–25 minutes of arrival).

4.5 If no hemorrhage is identified and the patient meets criteria, CT angiography (arch to vertex) is obtained to identify large vessel occlusion and determine candidacy for mechanical thrombectomy.

4.6 If hemorrhage is identified, the patient is managed per the Hemorrhagic Stroke Management policy (SU-004) and neurosurgery is consulted.

5. Procedure — IV Thrombolytic Administration

5.1 Blood pressure must be <185/110 mmHg prior to thrombolytic administration; antihypertensive therapy (e.g., labetalol, nicardipine, or clevidipine per organizational protocol) is administered if needed. If blood pressure cannot be safely lowered below this threshold, thrombolytic therapy is withheld.

5.2 Alteplase is dosed at 0.9 mg/kg (maximum total dose 90 mg): 10% administered as an IV bolus over 1 minute, followed by the remaining 90% infused over 60 minutes. If tenecteplase is selected under the approved institutional acute ischemic stroke protocol, it is administered at 0.25 mg/kg IV (maximum 25 mg) as a single bolus. Agent selection follows the current AHA/ASA guideline, the approved hospital order set, and pharmacy policy.

5.3 Both alteplase and tenecteplase are appropriate IV thrombolytic options within the 4.5-hour window from LKW. Eligibility for the 3–4.5-hour portion of this window is determined using current AHA/ASA inclusion/exclusion criteria and physician judgment; advanced age (>80), oral anticoagulant use, high baseline NIHSS (>25), a combined history of prior stroke and diabetes, and extensive early ischemic change (>1/3 MCA territory) are factors that inform — but under the 2026 guideline no longer function as a fixed checklist of standing exclusions for — this window. 5.3.1 Imaging-selected IV thrombolysis: for patients with unknown time of onset (including wake-up stroke) or presenting 4.5–9 hours from LKW, IV thrombolysis may be offered when advanced imaging (perfusion or MRI-based mismatch, or DWI-FLAIR mismatch for unknown-onset patients) identifies salvageable tissue, per the 2026 AHA/ASA guideline and institutional imaging-selection protocol.

5.4 During and after infusion, the patient is monitored in a high-dependency setting (ICU, Stroke Unit, or ED resuscitation area) with neurological checks and blood pressure monitoring every 15 minutes for 2 hours, then every 30 minutes for 6 hours, then hourly for 16 hours.

5.5 Blood pressure is maintained <180/105 mmHg for at least the first 24 hours post-thrombolysis.

5.6 No antiplatelet or anticoagulant agents are administered for 24 hours following thrombolysis, pending a follow-up non-contrast CT to exclude hemorrhagic transformation.

5.7 Any acute neurological deterioration, severe headache, new hypertension, nausea, or vomiting during or after infusion triggers immediate cessation of the infusion (if ongoing), STAT non-contrast head CT, and physician notification to evaluate for symptomatic intracranial hemorrhage.

6. Procedure — Mechanical Thrombectomy

6.1 Patients with confirmed anterior circulation LVO, NIHSS ≥6, pre-stroke independent functional status, and favorable imaging profile (e.g., ASPECTS ≥6) are evaluated for mechanical thrombectomy up to 24 hours from LKW using clinical-imaging mismatch criteria (CT perfusion or MRI-based core/penumbra mismatch) for the 6–24-hour window. Per the 2026 AHA/ASA guideline, selected patients with a large ischemic core (low ASPECTS/larger core volume on CT perfusion or MRI) are also evaluated for thrombectomy rather than excluded by imaging volume alone, based on individualized assessment of core size, clinical severity, and time from onset. Basilar artery occlusion: thrombectomy is strongly recommended for appropriately selected patients with basilar artery occlusion within 24 hours of LKW, particularly those with NIHSS ≥10, per the 2026 AHA/ASA guideline; case-by-case evaluation with Neurointervention/Neurosurgery continues to apply for patients outside this profile.

6.2 The neurointerventionalist/neurosurgery team is notified immediately upon identification of a thrombectomy candidate; if not available on-site, an emergent transfer to the designated thrombectomy-capable partner facility is initiated per the organization's stroke transfer agreement.

6.3 IV thrombolysis (if the patient is eligible and within window) is not withheld while arranging or performing mechanical thrombectomy.

6.4 Post-procedure patients are admitted to the ICU or Stroke Unit for neurological and access-site monitoring.

7. Procedure — Extended Window Evaluation (6–24 Hours)

7.1 Patients presenting 6–24 hours from LKW are screened using BE-FAST; a positive screen prompts CT stroke protocol (non-contrast CT, CTA, CT perfusion).

7.2 Clinical-imaging mismatch is assessed using validated automated perfusion software, applying core infarct volume and NIHSS thresholds consistent with the pivotal randomized trial criteria referenced in current AHA/ASA guidelines.

7.3 Patients meeting mismatch criteria are evaluated for mechanical thrombectomy per Section 6.

8. General Supportive Care (All Acute Ischemic Stroke Patients)

8.1 Supplemental oxygen to maintain $SpO_2 \geq 94\%$.

8.2 Continuous cardiac monitoring for at least the first 48 hours to detect atrial fibrillation and other arrhythmias.

8.3 Treat hyperthermia (temperature >38°C) with antipyretics and identify the underlying source.

8.4 Correct hypovolemia with isotonic IV fluids; avoid fluid overload in patients with reduced ejection fraction.

8.5 Permissive hypertension in patients not receiving thrombolysis/thrombectomy: withhold antihypertensive therapy unless systolic BP >220 mmHg or diastolic BP >120 mmHg, or unless a compelling comorbid indication exists (e.g., acute coronary syndrome, aortic dissection, hypertensive encephalopathy); when treatment is indicated, lower BP by approximately 15% during the first 24 hours.

8.6 Initiate VTE prophylaxis (intermittent pneumatic compression ± prophylactic-dose subcutaneous anticoagulation per institutional VTE prophylaxis policy) as early as clinically appropriate.

8.7 Initiate aspirin within 24–48 hours of stroke onset in patients not receiving thrombolysis, or 24 hours after thrombolysis once hemorrhage is excluded, unless contraindicated.

8.8 Statin therapy is initiated or continued during hospitalization per current lipid management guidelines.

8.9 Dysphagia screening is completed prior to any oral intake, per the NIHSS/Dysphagia Screening policy (SU-006).

9. Admission Criteria

Patients are admitted to ICU, Stroke Unit, or a monitored bed based on treatment received and clinical status, per the criteria below:

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-003	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Medical Director		
APPLIES TO	Hospital Wide		

9.1 ICU/high-dependency admission: patients who received IV thrombolysis and/or mechanical thrombectomy; acute stroke with depressed consciousness; stroke with seizures; hemodynamic or respiratory instability.

9.2 Stroke Unit admission: hemodynamically stable acute ischemic stroke or TIA not requiring ICU-level monitoring, per the Stroke Unit Admission/Discharge Criteria in policy SU-001/SU-005.

9.3 Any patient with new focal deficit, GCS decline, need for airway protection, or hemorrhagic transformation is escalated to ICU immediately.

Scope clarification — Adult Stroke Program: SU-003 and SU-014 apply to patients aged 18 years and older. King Fahad Central Hospital does not provide pediatric stroke care under this Stroke Program. Patients younger than 18 years with suspected stroke are outside the scope of this manual and are managed or transferred according to the hospital pediatric referral and transfer policy. Adult IV thrombolysis and mechanical thrombectomy criteria in this manual must not be applied to pediatric patients.

10. References

10.1 Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association. Stroke. doi:10.1161/STR.0000000000000513.

10.2 AHA/ASA Target: Stroke® Best Practice Strategies.

10.3 American Stroke Association International Comprehensive Stroke Center Certification Standards.

11. Attachments

11.1 IV Thrombolytic Inclusion/Exclusion Checklist.

11.2 Thrombolytic Therapy (Alteplase/Tenecteplase) Administration and Monitoring Form.

11.3 Post-IV Thrombolysis Neurological and Vital Sign Monitoring Flow Sheet.

11.4 Mechanical Thrombectomy Transfer Checklist (if applicable).

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-004	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Medical Director		
APPLIES TO	Emergency Department, ICU, Stroke Unit, Neurosurgery		

SU-004 Spontaneous Intracerebral and Subarachnoid Hemorrhage Management

1. Purpose

To standardize the emergent evaluation, blood pressure management, reversal of anticoagulation, and neurosurgical coordination for patients with spontaneous intracerebral hemorrhage (ICH) or subarachnoid hemorrhage (SAH), consistent with AHA/ASA guidelines for the management of spontaneous intracerebral hemorrhage and aneurysmal subarachnoid hemorrhage.

2. Definitions

2.1 ICH: Intracerebral (parenchymal) hemorrhage.

2.2 SAH: Subarachnoid hemorrhage.

2.3 AVM: Arteriovenous malformation.

2.4 ICP: Intracranial pressure.

2.5 EVD: External ventricular drain.

3. Policy

3.1 Any hemorrhage identified on initial CT during Stroke Code evaluation triggers immediate neurosurgical notification and CT angiography of the Circle of Willis to evaluate for a secondary vascular cause (AVM, aneurysm, arteriovenous fistula).

3.2 Patients with ICH or SAH are managed in the ICU with continuous neurological and hemodynamic monitoring.

3.3 Anticoagulant reversal, when indicated, is initiated emergently without waiting for transfer or definitive imaging, per the organizational anticoagulation reversal protocol.

3.4 Blood pressure management follows AHA/ASA ICH guideline targets to reduce hematoma expansion risk while maintaining cerebral perfusion.

4. Procedure — Intracerebral Hemorrhage

4.1 Confirm ICH on non-contrast CT; obtain CT angiography (arch to vertex / Circle of Willis) to evaluate for an underlying structural lesion, particularly in younger patients, those without a history of hypertension, or with atypical hemorrhage location.

4.2 Notify neurosurgery immediately for evaluation of surgical candidacy (hematoma evacuation, ventriculostomy) and level of care.

4.3 Blood pressure management: for patients presenting with systolic BP 150–220 mmHg, acute lowering to a target systolic BP of approximately 140 mmHg is reasonable and can be safe, using IV agents such as labetalol, nicardipine, or clevidipine. SBP is generally maintained in the range of approximately 130–150 mmHg; lowering SBP below 130 mmHg is avoided, as it may be associated with harm. Targets are reassessed based on individual clinical context.

4.4 Reverse anticoagulation emergently in patients on vitamin K antagonists (e.g., with 4-factor prothrombin complex concentrate and IV vitamin K), direct oral anticoagulants (with specific reversal agents where available, or PCC), or heparin (protamine sulfate), per the organizational reversal protocol.

4.5 Correct severe thrombocytopenia or coagulation factor deficiencies as clinically indicated; platelet transfusion is not routinely recommended for antiplatelet-associated ICH without a specific procedural indication.

4.6 Manage elevated ICP as needed: head-of-bed elevation to 30°, analgesia/sedation as needed, osmotic therapy (mannitol or hypertonic saline) for signs of herniation or elevated ICP, and ventriculostomy for hydrocephalus per neurosurgical guidance.

4.7 Maintain normoglycemia, normothermia, and seizure precautions; prophylactic anticonvulsants are not routinely recommended in the absence of clinical or electrographic seizures.

4.8 VTE prophylaxis with intermittent pneumatic compression is initiated on admission; pharmacologic prophylaxis may be added after documented cessation of bleeding, typically 24–48 hours after onset, per physician judgment.

4.9 Decompressive craniectomy ± hematoma evacuation is considered for select patients with large hemorrhage and mass effect or clinical deterioration, per neurosurgical evaluation.

5. Procedure — Subarachnoid Hemorrhage

5.1 Suspected SAH (thunderclap headache, meningismus) with a non-diagnostic non-contrast CT is further evaluated with lumbar puncture (evaluating for xanthochromia) if clinically indicated and CT is negative but suspicion remains high.

5.2 Confirmed SAH triggers emergent CT angiography and neurosurgical/neurointerventional consultation for aneurysm identification and securement (surgical clipping or endovascular coiling), preferably within 24 hours of symptom onset, consistent with AHA/ASA 2023 guidance for the management of aneurysmal subarachnoid hemorrhage.

5.3 Nimodipine 60 mg PO/NG every 4 hours is initiated for vasospasm prophylaxis, per protocol, unless contraindicated by hypotension.

5.4 Blood pressure is controlled to reduce rebleeding risk prior to aneurysm securement, balanced against maintaining adequate cerebral perfusion; specific targets are individualized with neurosurgery/neurocritical care.

5.5 Patients are monitored for delayed cerebral ischemia (vasospasm) with serial neurological assessments and, where available, transcranial Doppler monitoring, typically most pronounced days 4–14 post-hemorrhage.

5.6 Hydrocephalus is managed with external ventricular drainage per neurosurgical assessment.

5.7 Euvolemia is maintained; prophylactic hypervolemic or induced hypertensive therapy is not used routinely but may be considered for symptomatic vasospasm under neurocritical care guidance.

6. Admission and Escalation Criteria

6.1 All ICH and SAH patients are admitted to the ICU.

6.2 Escalation to neurosurgical intervention is triggered by: GCS decline ≥2 points, new anisocoria, hematoma expansion on repeat imaging, hydrocephalus, or signs of herniation.

7. References

7.1 Greenberg SM, et al. 2022 AHA/ASA Guideline for the Management of Patients with Spontaneous Intracerebral Hemorrhage. Stroke.

7.2 Hoh BL, et al. 2023 Guideline for the Management of Patients With Aneurysmal Subarachnoid Hemorrhage: A Guideline From the American Heart Association/American Stroke Association. Stroke. doi:10.1161/STR.0000000000000436.

7.3 American Stroke Association International Comprehensive Stroke Center Certification Standards.

8. Attachments

8.1 Anticoagulation Reversal Protocol.

8.2 ICH Blood Pressure Management Order Set.

8.3 SAH / Vasospasm Monitoring Flow Sheet.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-005	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Medical Director		
APPLIES TO	Emergency Department, Stroke Unit, Outpatient Neurology Clinic		

SU-005 Transient Ischemic Attack (TIA) Evaluation and Management

1. Purpose

To standardize the risk stratification, urgent evaluation, and secondary prevention management of patients presenting with transient ischemic attack (TIA), given the substantial short-term risk of subsequent stroke.

2. Definitions

2.1 TIA: A transient episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia, without evidence of acute infarction on imaging.

2.2 $ABCD^2$ Score: A clinical risk-stratification tool for early stroke risk after TIA (Age, Blood pressure, Clinical features, Duration of symptoms, Diabetes).

2.3 Crescendo TIA: Recurrent TIAs with increasing frequency, duration, or severity over a short period.

3. Policy

3.1 All patients presenting with suspected TIA undergo the same urgent Stroke Code evaluation pathway as acute ischemic stroke (SU-002), because clinical differentiation between TIA and acute ischemic stroke cannot reliably be made on presentation.

3.2 Patients with TIA are at significantly increased short-term risk of stroke (up to approximately 10–20% within 90 days, with the highest risk in the first 48 hours), warranting urgent risk stratification and initiation of secondary prevention.

3.3 High-risk TIA patients (including crescendo TIA, $ABCD^2$ ≥4, or TIA due to identified large-vessel disease/atrial fibrillation) are admitted for expedited work-up; lower-risk patients may be managed through an expedited outpatient TIA pathway when a same-day/next-day evaluation can be reliably arranged.

4. Procedure

4.1 Perform BE-FAST/NIHSS assessment and non-contrast head CT (or MRI with diffusion-weighted imaging where readily available) to exclude acute infarction or hemorrhage.

4.2 Calculate and document the $ABCD^2$ score to support admission/disposition decision-making, used in conjunction with overall clinical judgment.

4.3 Obtain vascular imaging (CT angiography or MR angiography of the head and neck) to evaluate for large-vessel stenosis or occlusion.

4.4 Obtain cardiac evaluation: 12-lead ECG on arrival, and continuous cardiac (telemetry) monitoring for at least 24–48 hours to detect paroxysmal atrial fibrillation; extended outpatient cardiac monitoring is arranged for patients with cryptogenic events and no arrhythmia identified during admission.

4.5 Obtain echocardiography (transthoracic, with transesophageal as indicated) in patients with suspected cardioembolic source, particularly those <60 years or with cryptogenic TIA.

4.6 Initiate antithrombotic therapy per indication rather than a single default duration: aspirin monotherapy for standard-risk TIA/minor stroke; short-term dual antiplatelet therapy (aspirin plus clopidogrel) for 21 days in patients with minor ischemic stroke (NIHSS ≤3) or high-risk TIA ($ABCD^2$ ≥4) presenting within 24 hours of onset; aspirin plus ticagrelor for up to 30 days may be considered in selected minor stroke/high-risk TIA patients per current AHA/ASA guidance. Anticoagulation is used instead of antiplatelet therapy when a cardioembolic source is identified.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-005	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Medical Director		
APPLIES TO	Emergency Department, Stroke Unit, Outpatient Neurology Clinic		

4.7 Initiate or optimize statin therapy and blood pressure management per secondary prevention guidelines.

4.8 For patients with symptomatic carotid stenosis ≥70% (or 50–69% with additional risk factors), refer promptly for carotid revascularization evaluation (carotid endarterectomy or carotid artery stenting), ideally within 2 weeks of the event.

4.9 Provide stroke risk-factor education and secondary prevention counseling prior to discharge, per policy SU-008.

4.10 Arrange follow-up with neurology within 7 days of discharge for all TIA patients managed as outpatients or discharged from observation.

5. Admission Criteria

5.1 Crescendo or recurrent TIA.

5.2 ABCD² score ≥4.

5.3 Identified symptomatic large-vessel stenosis or intracranial/extracranial occlusion.

5.4 Suspected cardioembolic source requiring further work-up.

5.5 Inability to reliably complete urgent outpatient evaluation within 24–48 hours.

6. References

6.1 Kleindorfer DO, et al. 2021 AHA/ASA Guideline for the Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack. Stroke.

6.2 Johnston SC, et al. Validation and refinement of the ABCD² score.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-006	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Program Coordinator / Medical Director		
APPLIES TO	Emergency Department, Stroke Unit, ICU		

SU-006 NIHSS Assessment, Neurological Monitoring and Dysphagia/Swallow Screening

1. Purpose

To standardize neurological assessment using the NIH Stroke Scale (NIHSS) and to ensure all stroke patients are screened for dysphagia prior to any oral intake, reducing the risk of aspiration pneumonia — a core AHA/ASA and American Stroke Association certification stroke performance measure.

2. Policy — NIHSS

2.1 All physicians and registered nurses performing stroke assessments must be NIHSS-certified prior to independently scoring the scale, with recertification per organizational policy.

2.2 NIHSS is documented at initial presentation, and at defined intervals thereafter (e.g., 2 hours and 24 hours post-thrombolysis, and with any acute neurological change) to detect early deterioration or improvement.

2.3 A sustained increase in NIHSS of ≥4 points, or any acute decline in level of consciousness, triggers immediate physician notification and evaluation, including consideration of an urgent non-contrast head CT to exclude hemorrhagic transformation or infarct progression.

3. Policy — Dysphagia / Swallow Screening

3.1 All acute stroke patients remain nil per oral (NPO), including oral medications, until a formal swallow screen is completed and passed.

3.2 A validated swallow screening tool (e.g., a standardized water-swallow protocol) is administered by an appropriately trained clinician (typically a trained nurse) within 4 hours of arrival, and prior to any oral intake of food, fluid, or medication.

3.3 Swallow screening is deferred only for patients who are NPO for other reasons (e.g., planned intervention) or who are not alert enough to participate; screening is completed as soon as the patient is alert and able to participate.

3.4 Patients who fail the screening remain NPO and are referred to a speech-language pathologist for comprehensive assessment, to be completed within 24 hours of referral.

3.5 Instrumental evaluation (fiberoptic endoscopic evaluation of swallowing [FEES] or videofluoroscopic swallow study) is performed when needed to confirm the presence/absence of aspiration and guide the treatment plan.

3.6 Patients with confirmed dysphagia receive individualized dietary modification, compensatory swallowing strategies, and swallowing exercises as appropriate.

3.7 Patients who fail repeat swallow assessment or require prolonged NPO status are considered for enteral nutrition access (nasogastric, nasoduodenal, or percutaneous endoscopic gastrostomy) to maintain nutrition and hydration.

3.8 Swallow screening is repeated whenever a patient's neurological or medical status changes.

4. Documentation

NIHSS scores and swallow screening results (pass/fail, time completed, screening tool used) are documented in the medical record and are subject to concurrent and retrospective audit as part of the stroke performance improvement program (SU-009), consistent with GWTG-Stroke/American Stroke Association certification measure STK-1 (dysphagia screening) requirements.

5. References

5.1 National Institutes of Health Stroke Scale (NIHSS) Training and Certification materials.

5.2 AHA/ASA Get with the Guidelines®–Stroke Dysphagia Screening Measure Specifications.

5.3 American Stroke Association stroke quality measures and applicable International Stroke Center Certification requirements.

6. Attachments

6.1 NIHSS Scoring Form.

6.2 Bedside Swallow Screening Tool.

6.3 Speech-Language Pathology Comprehensive Dysphagia Assessment Form.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-007	—	]Effective Date[	]Review Date[
APPROVED BY	Physiatrist / Medical Director		
APPLIES TO	Stroke Unit, ICU, Rehabilitation Services		

SU-007 Stroke Rehabilitation and Early Mobilization

1. Purpose

To ensure timely, structured, multidisciplinary rehabilitation assessment and intervention for all stroke patients, consistent with AHA/ASA guidelines for stroke rehabilitation and recovery, in order to optimize functional outcomes.

2. Policy

2.1 All patients admitted with a diagnosis of stroke receive a physiatry/rehabilitation medicine consultation within 24 hours of admission.

2.2 All patients undergo a full multidisciplinary rehabilitation needs assessment (physical therapy, occupational therapy, speech-language pathology, as indicated), documented in the multidisciplinary plan of care.

2.3 Early mobilization is initiated within 24 hours of stroke onset for medically stable patients, by an appropriately trained healthcare professional with access to appropriate mobility equipment; very early (within 24 hours), high-frequency, high-dose out-of-bed mobilization is avoided per trial evidence, and mobilization intensity is individualized.

2.4 A discharge rehabilitation disposition (home with outpatient therapy, home health, inpatient rehabilitation facility, or skilled nursing facility) is determined by the multidisciplinary team prior to discharge, based on functional assessment.

3. Procedure — Rehabilitation Assessment

3.1 The physiatrist assesses the patient and documents individualized goals and a plan of care in the Physical Medicine Request Form and multidisciplinary plan of care form.

3.2 Assessment addresses: motor function (strength, tone, coordination), sensory function, swallowing (per SU-006), bladder and bowel function, skin integrity/pressure injury risk, cognition/communication, vision/hearing, nutritional status, and psychosocial needs.

3.3 A skin/pressure-injury prevention and positioning program is established, addressing contracture prevention, respiratory function, and use of specialized equipment (splints, braces) as needed.

4. Procedure — Therapeutic Interventions

4.1 Physical and occupational therapists implement a motor control/retraining program including facilitation/inhibition techniques, stretching, biofeedback, therapeutic exercise, and neuromuscular electrical stimulation as clinically indicated.

4.2 Functional mobility training addresses: bed mobility, transfers (bed, chair, wheelchair), stair negotiation, ambulation with or without assistive devices, and wheelchair mobility/propulsion, individualized to the patient's deficits and goals.

4.3 Occupational therapy addresses activities of daily living, upper-extremity function, cognitive-perceptual deficits, and home/environmental modification needs.

4.4 Speech-language pathology addresses communication deficits (aphasia, dysarthria) and cognitive-communication impairments in addition to dysphagia management (SU-006).

4.5 Psychological/neuropsychological support is arranged for patients with post-stroke depression, anxiety, or cognitive impairment affecting rehabilitation participation.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-007	—	]Effective Date[	]Review Date[
APPROVED BY	Physiatrist / Medical Director		
APPLIES TO	Stroke Unit, ICU, Rehabilitation Services		

5. Discharge Rehabilitation Planning

5.1 The multidisciplinary team, in coordination with Case Management, determines the appropriate level of post-acute rehabilitation care based on functional status, medical stability, cognitive capacity to participate in therapy, and caregiver/family support.

5.2 All stroke patients are formally assessed for and considered for rehabilitation services prior to discharge, and this assessment is documented, consistent with the GWTG-Stroke/American Stroke Association certification rehabilitation assessment performance measure.

6. References

6.1 Winstein CJ, et al. AHA/ASA Guidelines for Adult Stroke Rehabilitation and Recovery. Stroke.

6.2 AHA/ASA Get with the Guidelines®–Stroke Rehabilitation Assessment Measure Specifications.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-008	—	]Effective Date[	]Review Date[
APPROVED BY	Medical Director / Stroke Program Coordinator		
APPLIES TO	Stroke Unit, ICU, Case Management		

SU-008 Discharge Planning, Secondary Prevention and Patient/Family Education

1. Purpose

To ensure that every stroke and TIA patient is discharged with an evidence-based secondary prevention regimen, appropriate follow-up, and documented patient/family education, consistent with AHA/ASA secondary prevention guidelines and Joint Commission stroke discharge performance measures (STK set).

2. Policy

2.1 Discharge planning begins at admission and is reassessed daily by the multidisciplinary team.

2.2 Prior to discharge, every eligible ischemic stroke/TIA patient is prescribed antithrombotic therapy (antiplatelet or anticoagulant, as clinically indicated) unless contraindicated and documented.

2.3 Patients with atrial fibrillation/flutter are prescribed anticoagulation at discharge unless contraindicated and documented.

2.4 Patients with LDL ≥100 mg/dL, or with atherosclerotic disease regardless of LDL, are prescribed statin therapy at discharge unless contraindicated.

2.5 Smoking cessation counseling and resources are provided to all current smokers.

2.6 Formal stroke education is provided to the patient and/or caregiver, covering the required core topics in Section 3.

2.7 A stroke rehabilitation needs assessment is documented prior to discharge (SU-007).

2.8 Follow-up appointment(s) are scheduled prior to discharge, including neurology follow-up within 7–14 days as clinically appropriate.

3. Required Patient/Family Education Topics

3.1 Personal risk factors for stroke and individualized modification strategies (hypertension, diabetes, dyslipidemia, atrial fibrillation, smoking, physical inactivity, obesity).

3.2 Warning signs and symptoms of stroke (BE-FAST) and the importance of calling emergency services immediately if symptoms recur.

3.3 Prescribed medications: name, purpose, dose, schedule, and importance of adherence, including anticoagulant/antiplatelet-specific precautions.

3.4 Follow-up care: scheduled appointments, and who to contact with questions or concerns.

3.5 Activity, dietary, and lifestyle recommendations, including dysphagia diet modifications if applicable.

3.6 Community resources, support groups, and caregiver support services.

4. Discharge Criteria from the Stroke Unit

Discharge or transfer from the Stroke Unit is at the discretion of the treating physician in consultation with the referring physician/department, guided by the following:

4.1 The patient is neurologically and hemodynamically stable, with status continuously reassessed to identify patients who no longer require Stroke Unit–level care.

4.2 The patient's physiologic status has stabilized such that intensive monitoring is no longer required, or no further benefit is expected from continued Stroke Unit stay.

4.3 The etiology of the stroke has been identified, or a clear plan for completion of the diagnostic work-up (inpatient or outpatient) has been established.

4.4 If physiologic status has deteriorated and further active intervention is not planned (e.g., transition to comfort-focused care), transfer to a lower level of care is appropriate.

4.5 Disposition (home, home with services, inpatient rehabilitation, skilled nursing facility) is finalized in coordination with Case Management/Social Work and the patient/family.

5. Documentation

Discharge medication reconciliation, education topics covered, and follow-up plans are documented in the discharge summary and nursing discharge instructions, and are subject to audit under the stroke performance improvement program (SU-009).

6. References

6.1 Kleindorfer DO, et al. 2021 AHA/ASA Guideline for the Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack.

6.2 AHA/ASA Get with the Guidelines®–Stroke Discharge Measure Specifications (STK-1 through STK-10).

6.3 American Stroke Association International Stroke Center Certification quality and performance requirements.

7. Attachments

7.1 Stroke Discharge Checklist.

7.2 Patient/Family Stroke Education Booklet and Documentation Form.

7.3 Discharge Medication Reconciliation Form.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-009	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

SU-009 Stroke Performance Improvement, Data Registry and Peer Review

1. Purpose

To establish a continuous, data-driven performance improvement process for the Stroke Program, including participation in a standardized national stroke registry, in fulfillment of AHA/ASA Get with the Guidelines®–Stroke and Joint Commission stroke certification requirements.

2. Policy

2.1 All patients with a confirmed diagnosis of ischemic stroke, hemorrhagic stroke, TIA, or subarachnoid hemorrhage are entered into the stroke data registry by the Stroke Program Coordinator/data abstractor.

2.2 Required standardized performance measures are tracked continuously, submitted through the applicable stroke registry/QCT as required, and reviewed by the Stroke Committee at minimum quarterly. For Comprehensive Stroke Center (CSC) readiness, the program shall use the measure names and sequence in the American Stroke Association International Stroke Center Certification Program Manual (Version 2.0):

2.2.1 Primary Stroke Center & Comprehensive Stroke Center Performance Measures (Measures 1–8):
1. Venous Thromboembolism (VTE): ischemic or hemorrhagic stroke patients who received VTE prophylaxis, or have documentation why prophylaxis was not given, on the day of or the day after hospital admission.
2. Discharged on Antithrombotic Therapy: ischemic stroke patients prescribed antithrombotic therapy at hospital discharge.
3. Anticoagulation Therapy for Atrial Fibrillation/Flutter: ischemic stroke patients with atrial fibrillation/flutter prescribed anticoagulation therapy at hospital discharge.
4. Thrombolytic Therapy: acute ischemic stroke patients who arrive within the applicable measure time window and for whom IV thrombolytic therapy is initiated within the applicable treatment window, according to the current Association measure specification.
5. Antithrombotic Therapy by End of Hospital Day Two: ischemic stroke patients administered antithrombotic therapy by the end of hospital day 2.
6. Discharged on Statin Medication: ischemic stroke patients prescribed statin medication at hospital discharge.
7. Stroke Education: ischemic or hemorrhagic stroke patients or caregivers provided stroke education addressing emergency activation, follow-up after discharge, discharge medications, patient-specific stroke risk factors, and stroke warning signs/symptoms.
8. Assessed for Rehabilitation: ischemic or hemorrhagic stroke patients assessed for rehabilitation services.

2.2.2 Comprehensive Stroke Center Performance Measures (Measures 9–16):
9. National Institutes of Health Stroke Scale (NIHSS Score Performed for Ischemic Stroke Patients): initial NIHSS documented before acute recanalization therapy for treated patients, or within 12 hours of emergency department arrival for patients not undergoing recanalization therapy.
10. Hemorrhagic Transformation (Overall Rate): symptomatic intracranial hemorrhage within 36 hours after IV thrombolytic, intra-arterial thrombolytic, or mechanical endovascular reperfusion treatment, using the Association measure definition.
11. Hemorrhagic Transformation for Patients Treated with Intravenous (IV) Thrombolytic Therapy Only: symptomatic intracranial hemorrhage within 36 hours after IV thrombolytic therapy only, using the Association measure definition.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-009	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

12. Hemorrhagic Transformation for Patients Treated with Intra-Arterial (IA) Thrombolytic Therapy or Mechanical Endovascular Reperfusion Therapy: symptomatic intracranial hemorrhage within 36 hours after IA thrombolytic or mechanical endovascular reperfusion therapy, using the Association measure definition.
13. Thrombolysis in Cerebral Infarction (TICI Post-Treatment Reperfusion Grade): post-treatment TICI grade 2B or higher at the end of IA thrombolytic and/or mechanical endovascular reperfusion therapy.
14. Modified Rankin Score (mRS at 90 Days: Favorable Outcome): treated ischemic stroke patients with mRS ≤2 at 90 days, using the Association assessment window of 75–105 days.
15. Timeliness of Reperfusion — Arrival Time to TICI 2B or Higher: eligible large-vessel-occlusion patients receiving mechanical endovascular reperfusion therapy and achieving TICI 2B or higher within the Association-defined arrival-to-reperfusion interval (≤150 minutes in the current Program Manual).
16. Timeliness of Reperfusion — Skin Puncture to TICI 2B or Higher: eligible large-vessel-occlusion patients receiving mechanical endovascular reperfusion therapy and achieving TICI 2B or higher within ≤60 minutes of skin puncture.
2.2.3 Optional CSC Measures (Measures 17–20), tracked when adopted by the Stroke Program:
17. Severity Measurement Performed for SAH and ICH Patients (Overall Rate): Hunt and Hess Scale for SAH or ICH Score for ICH documented before surgical intervention, or within 6 hours of hospital arrival for patients not undergoing surgical intervention.
18. Hunt and Hess Scale Performed for SAH Patients: documented before surgical intervention, or within 6 hours of hospital arrival when no surgical intervention is performed.
19. ICH Score Performed for ICH Patients: documented before surgical intervention, or within 6 hours of hospital arrival when no surgical intervention is performed.
20. Nimodipine Treatment Administered: SAH patients receiving nimodipine within 24 hours of hospital arrival.
2.2.4 Mandatory CSC Metric:
21. Arrival Time to Skin Puncture: median time, in minutes, from hospital arrival to arterial skin puncture for acute ischemic stroke patients undergoing endovascular treatment.
2.2.5 Additional institutional process indicators (e.g., door-to-CT, door-to-needle, stroke-code response, complication rates, mortality, length of stay, and other locally selected indicators) may be monitored for quality improvement but shall be displayed separately from the Association Required Standardized Measures to avoid misclassification.
2.2.6 Measure definitions, numerator/denominator logic, exclusions, treatment windows, and submission specifications shall follow the current American Stroke Association International Stroke Center Certification Program Manual and the applicable registry/QCT specifications. If an Association specification is revised, the current specification supersedes any conflicting wording in this policy.

2.3 Cases with a protocol deviation, unexpected complication, or mortality undergo structured peer review (Stroke Morbidity & Mortality review), with findings reported to the Stroke Committee and, as applicable, the Medical Staff Quality/Peer Review Committee.

2.4 Performance below internal or national benchmark thresholds triggers a documented corrective action plan with assigned responsibility, timeline, and re-measurement.

2.5 Aggregate and case-level data are used to inform staff education, protocol revision, and resource allocation decisions.

2.6 An annual Stroke Program performance report is prepared summarizing volumes, outcome metrics, and quality improvement initiatives, for review by hospital leadership and the Medical Staff.

3. Procedure

3.1 Concurrent case identification: stroke/TIA cases are flagged in real time via the Stroke Code activation log, ED diagnosis coding, and daily census review by the Stroke Coordinator.

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SU-009	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

3.2 Data abstraction: the Stroke Coordinator abstracts required data elements into the registry (e.g., GWTG-Stroke) within the timeframe specified by the registry's data completion requirements.

3.3 Quarterly Stroke Committee review of aggregate performance measures, benchmarked against AHA/ASA and American Stroke Association certification thresholds.

3.4 Case-level review of all thrombolytic/thrombectomy complications, in-hospital stroke mortality, and any case flagged by nursing, physician, or risk management referral.

3.5 Action plans arising from performance gaps are tracked to closure and re-audited at the subsequent Committee meeting.

4. References

4.1 AHA/ASA Get with the Guidelines®–Stroke Program Manual and Measure Specifications.

4.2 American Stroke Association International Stroke Center Certification Program Manual, Version 2.0 (2024), Required Standardized Measures, pp. 16–17; current QCT/registry measure specifications.

5. Attachments

5.1 Stroke Registry Data Abstraction Tool.

5.2 Stroke Morbidity & Mortality Review Form.

5.3 Quarterly Stroke Committee Performance Dashboard Template.

POLICY NUMBER	REPLACES NUMBER	EFFECTIVE DATE	REVIEW DATE
SU-010	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

SU-010 Code Stroke Team — Roles, Responsibilities and Activation Roster

1. Purpose

To define the composition, individual roles, and activation roster of the Code Stroke Team, and to ensure that every member understands their specific responsibilities during a Stroke Code so that the team functions as a coordinated unit from activation through disposition.

2. Scope

This policy applies to all physicians, nurses, technologists, pharmacists, and allied staff who are designated members of the Code Stroke Team, and to any staff member responsible for activating or responding to a Stroke Code, in any hospital location (Emergency Department, inpatient units, ICU, Radiology, or off-site areas).

3. Definitions and Abbreviations

3.1 Code Stroke Team: The designated multidisciplinary group that responds immediately to a Stroke Code activation, as distinct from the broader Multidisciplinary Stroke Team described in SU-001, which also includes rehabilitation and longitudinal care disciplines.

3.2 Team Leader: The physician (Emergency Physician or Neurologist, per institutional assignment) who directs the Code Stroke response and has final authority for treatment decisions.

3.3 Recorder: The nurse or designated staff member responsible for real-time timestamping of all Stroke Code milestones on the Stroke Code Flow Sheet (Appendix C).

3.4 Runner: Support staff member who physically transports labs, blood products, or imaging orders when electronic/pneumatic systems are unavailable or delayed.

4. Policy

4.1 A designated Code Stroke Team responds to every Stroke Code activation (SU-002) within the defined time targets, 24 hours a day, 7 days a week, 365 days a year.

4.2 Each team role is filled by a specific position on every shift; role assignments are posted on the daily staffing assignment board and are not dependent on any single named individual.

4.3 Team members carry a dedicated Stroke Code pager/communication device or are reachable through the hospital's overhead/electronic paging system at all times while on duty.

4.4 All Code Stroke Team members complete role-specific competency validation upon assignment to the team and annually thereafter, in accordance with SU-015 (Nursing and Staff Stroke Education Program).

4.5 A debrief is conducted after every Stroke Code involving thrombolytic administration, a complication, or a significant time-target miss, to identify system issues and reinforce role performance.

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APPLIES TO	Hospital Wide		

5. Team Roles and Responsibilities

Role	Primary Responder	Key Responsibilities
Team Leader	Emergency Physician / Neurologist on duty	Directs the resuscitation; performs or supervises NIHSS; makes treatment eligibility decisions; reviews inclusion/exclusion checklist; obtains consent; communicates with family.
Stroke Team Nurse (Primary)	Stroke Unit / ED RN	Obtains vitals and IV access; draws STAT labs; prepares and administers thrombolytic per order; performs neuro-checks per monitoring schedule.
Recorder	Charge nurse or assigned RN	Timestamps every milestone on the Stroke Code Flow Sheet (activation, physician arrival, CT arrival/completion/read, needle time); ensures documentation is complete.
CT Technologist	Radiology on-call/on-duty tech	Prioritizes and immediately performs non-contrast CT ± CTA/CTP; notifies radiologist for urgent read.
Radiologist / Neuroradiologist	On-call Radiology	Provides urgent CT interpretation and communicates findings directly to the Team Leader by phone, not solely via written report.
Pharmacist	On-duty Pharmacy	Verifies weight-based thrombolytic dose; prepares medication on standby; double-checks dose prior to administration.
Laboratory Technologist	STAT lab on-duty	Prioritizes and expedites STAT stroke panel processing and result reporting.
Neurointerventionalist / Neurosurgery	On-call per roster	Evaluated emergently for large-vessel occlusion or hemorrhage requiring intervention (SU-003, SU-004, SU-014).
Charge Nurse (receiving unit)	Stroke Unit / ICU	Confirms bed availability and prepares for immediate patient transfer/admission.
Runner	Assigned support staff / aide	Physically expedites transport of specimens, blood products, or documents as needed.

6. Activation Roster and Contact Method

6.1 The current on-call roster for Neurology, Neurosurgery, and Neurointervention is maintained by the Stroke Program Coordinator and is available at all times through the hospital's on-call directory/paging system.

6.2 A single Stroke Code activation (call/page) simultaneously notifies all team roles listed in Section 5 through the hospital's group paging or clinical communication platform; no team member is notified sequentially or manually one-by-one.

6.3 If a team member does not acknowledge activation within 5 minutes, the charge nurse or Team Leader escalates through the on-call backup roster.

6.4 Team roster assignments (by role, not by name) are documented at the top of the Stroke Code Flow Sheet for every activation.

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7. References

7.1 AHA/ASA Target: Stroke® Best Practice Strategies — Organized Stroke Team Response.

7.2 American Stroke Association International Comprehensive Stroke Center Certification Standards.

8. Attachments

8.1 Code Stroke Team Roster and Role Assignment Board Template.

8.2 Stroke Code Flow Sheet (Appendix C).

8.3 Stroke Code Post-Event Debrief Form.

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APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

SU-011 Acute Stroke Immediate Management Pathway (0–24 Hour Clinical Pathway)

1. Purpose

To provide a single, consolidated, time-sequenced clinical pathway summarizing the immediate management of any patient presenting with suspected acute stroke, from the point of first contact through the first 24 hours, integrating the detailed policies referenced throughout this manual into one quick-reference sequence for the bedside team.

2. Scope

This pathway applies to all suspected acute stroke patients, pre-hospital and in-hospital, and is intended to be used alongside — not in place of — the detailed policies SU-002 (Stroke Code Activation), SU-003 (Acute Ischemic Stroke Management), and SU-004 (Hemorrhagic Stroke Management).

3. Policy

3.1 Every patient with suspected acute stroke moves through the pathway below without deviation unless a specific clinical contraindication is documented by the Team Leader.

3.2 Pathway time targets mirror the AHA/ASA Target: Stroke® goals defined in SU-002, Section 3.4.

3.3 Any step that cannot be completed within its target window is documented on the Stroke Code Flow Sheet with the reason for delay, for review by the Stroke Committee (SU-009).

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4. Immediate Management Pathway

MINUTE 0 — RECOGNITION
BE-FAST positive screen (pre-hospital or in-hospital) → ACTIVATE STROKE CODE

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MINUTES 0–10 — ARRIVAL / TRIAGE
Document exact Last Known Well (LKW) time • Vital signs + finger-stick glucose • IV access (18G preferred) • STAT labs drawn • Transfer to resuscitation/stroke bay • Team Leader begins focused history and NIHSS

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MINUTES 10–25 — IMAGING
Direct transport to CT • Non-contrast head CT ± CTA (arch to vertex) ± CTP • IV thrombolytic inclusion/exclusion checklist reviewed (Appendix B) • BP managed per algorithm (SU-012 / Appendix D or E)

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DECISION POINT — CT RESULT

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Hemorrhage identified	No hemorrhage — eligible for thrombolysis	No hemorrhage — not eligible / LVO suspected
Activate SU-004 pathway → Neurosurgery notified immediately → ICU admission → BP managed per hemorrhagic algorithm (Appendix E)	Proceed to thrombolytic administration within 60 minutes of arrival (goal ≤45 min) per SU-003 → monitor per post-tPA schedule	Obtain CTA/CTP if not already done → evaluate for mechanical thrombectomy (SU-014) up to 24 h from LKW

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MINUTES 25–60+ — TREATMENT
Thrombolytic administered and/or thrombectomy team activated • Continuous neuro and vital sign monitoring begins • Family/LAR informed and consent documented

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HOURS 1–24 — STABILIZATION AND ADMISSION
Admit to ICU or Stroke Unit per criteria (SU-017) • Repeat NIHSS at defined intervals • Dysphagia screen before any oral intake (SU-006) • VTE prophylaxis initiated • Continuous cardiac monitoring • Follow-up non-contrast CT at 24 h post-thrombolysis

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5. Escalation Triggers Throughout the Pathway

5.1 NIHSS increase ≥4 points or decline in level of consciousness — immediate physician notification and STAT CT.

5.2 Suspected symptomatic intracranial hemorrhage or systemic bleeding after thrombolytic — activate SU-013 Critical Actions immediately.

5.3 New large-vessel occlusion identified at any point — activate SU-014 Mechanical Thrombectomy Policy.

5.4 Any missed time target — document reason and notify the Stroke Coordinator for performance review (SU-009).

6. References

6.1 AHA/ASA 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke.

6.2 AHA/ASA Target: Stroke® Phase III Initiative.

7. Attachments

7.1 Stroke Code Flow Sheet, Illustrated (Appendix C).

7.2 IV Thrombolytic Inclusion/Exclusion Checklist (Appendix B).

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APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

SU-012 Blood Pressure Management in Acute Ischemic and Hemorrhagic Stroke

1. Purpose

To standardize blood pressure management targets and treatment algorithms for patients with acute ischemic stroke (with and without thrombolysis/thrombectomy) and for patients with acute hemorrhagic stroke (intracerebral or subarachnoid hemorrhage), consistent with current AHA/ASA guidelines.

2. Scope

This policy applies to all patients with confirmed or suspected acute stroke being managed in the Emergency Department, Stroke Unit, or ICU. It consolidates and standardizes the blood pressure provisions already referenced in SU-003 and SU-004 into a single dedicated policy with illustrated algorithms.

3. Policy — General Principles

3.1 Blood pressure targets differ fundamentally between ischemic and hemorrhagic stroke and by treatment received; the correct algorithm (Section 4 or 5) must be identified before any antihypertensive is administered.

3.2 Short-acting, titratable IV antihypertensive agents (labetalol, nicardipine, or clevidipine) are used in the acute phase; oral/long-acting agents are avoided until the patient is neurologically and hemodynamically stable.

3.3 Rapid, uncontrolled drops in blood pressure are avoided in all stroke subtypes, as hypotension can extend ischemic injury or reduce cerebral perfusion pressure.

3.4 All blood pressure treatment decisions and readings are documented on the Stroke Code Flow Sheet and subsequent nursing flow sheet at the frequency specified in Sections 4 and 5.

4. Algorithm — Acute Ischemic Stroke

STEP 1: Is the patient a candidate for IV thrombolysis and/or mechanical thrombectomy?

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YES — candidate for thrombolysis/thrombectomy	NO — not a candidate for acute reperfusion therapy
BEFORE treatment: lower BP to <185/110 mmHg using labetalol or nicardipine IV. If BP cannot be safely lowered below this threshold, thrombolysis is withheld. DURING/AFTER treatment (first 24 h): maintain BP <180/105 mmHg. Check BP every 15 min × 2 h, every 30 min × 6 h, then hourly × 16 h.	PERMISSIVE HYPERTENSION: withhold antihypertensive therapy unless SBP >220 mmHg or DBP >120 mmHg, or a compelling comorbid indication exists (ACS, aortic dissection, hypertensive encephalopathy, severe heart failure). If treatment is indicated, lower BP by approximately 15% during the first 24 hours — not to a normal target.

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Any BP elevation with acute neurological deterioration, severe headache, or new nausea/vomiting → STAT non-contrast head CT to exclude hemorrhagic transformation before further BP treatment

5. Algorithm — Acute Hemorrhagic Stroke (ICH / SAH)

STEP 1: Confirm hemorrhage subtype on non-contrast CT — Intracerebral Hemorrhage (ICH) or Subarachnoid Hemorrhage (SAH)

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Intracerebral Hemorrhage (ICH)	Subarachnoid Hemorrhage (SAH)
If SBP 150–220 mmHg: acute lowering to a target SBP of approximately 140 mmHg is reasonable and can be safe, using IV labetalol, nicardipine, or clevidipine; SBP is generally maintained ≈130–150 mmHg and SBP <130 mmHg is avoided. If SBP >220 mmHg: more aggressive reduction with continuous IV infusion and frequent (every 5–15 min) BP monitoring is reasonable, individualized to clinical context. Avoid precipitous drops; reassess and titrate continuously.	Control BP to reduce rebleeding risk prior to aneurysm securement (clipping/coiling), balanced against maintaining adequate cerebral perfusion pressure. Specific numeric targets are individualized with Neurosurgery/Neurocritical Care. Nimodipine 60 mg PO/NG every 4 hours is given for vasospasm prophylaxis (not for acute BP lowering) unless contraindicated by hypotension.

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All ICH/SAH patients: continuous IV antihypertensive infusion with automated BP monitoring every 5–15 minutes during active titration, transitioning to every 30–60 minutes once at target — managed in the ICU

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6. Documentation

Blood pressure readings, agent(s) used, and target achieved/not achieved are documented at the frequency specified above on the nursing flow sheet, and any deviation from target is flagged for physician reassessment.

7. References

7.1 Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association.

7.2 Greenberg SM, et al. 2022 AHA/ASA Guideline for the Management of Patients with Spontaneous Intracerebral Hemorrhage.

7.3 Hoh BL, et al. 2023 Guideline for the Management of Patients With Aneurysmal Subarachnoid Hemorrhage: A Guideline From the American Heart Association/American Stroke Association. Stroke. doi:10.1161/STR.0000000000000436.

8. Attachments

8.1 Blood Pressure Management Algorithm — Acute Ischemic Stroke, Illustrated (Appendix D).

8.2 Blood Pressure Management Algorithm — Acute Hemorrhagic Stroke, Illustrated (Appendix E).

8.3 IV Antihypertensive Agent Quick-Reference Dosing Card.

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APPLIES TO	Hospital Wide		

SU-013 Critical Actions — Suspected Intracranial Hemorrhage / Systemic Bleeding After IV Thrombolytic Therapy

1. Purpose

To provide an immediate, standardized set of critical actions for the bedside team when a patient develops signs of symptomatic intracranial hemorrhage or systemic bleeding during or after administration of an IV thrombolytic agent (alteplase/tenecteplase), minimizing time to recognition, imaging, reversal, and specialist notification.

2. Scope

This policy applies to any patient who has received or is receiving IV thrombolytic therapy for acute ischemic stroke, in any inpatient location, and to all nursing and physician staff caring for such patients.

3. Definitions

3.1 Symptomatic Intracranial Hemorrhage (sICH): Hemorrhage on imaging associated with a clinical deterioration of ≥4 points on the NIHSS, or any decline attributable to the hemorrhage.

3.2 Warning Signs: Any acute neurological deterioration, severe new headache, acute hypertension, nausea, or vomiting occurring during or within 24 hours of thrombolytic infusion.

3.3 Systemic Bleeding: Clinically significant bleeding from any site (gastrointestinal, genitourinary, IV access sites, oropharyngeal, retroperitoneal) occurring during or after thrombolytic administration.

4. Policy

4.1 Any nurse identifying a Warning Sign (Section 3.2) immediately stops the thrombolytic infusion if still running, without waiting for a physician order.

4.2 A STAT non-contrast head CT and STAT coagulation panel (including fibrinogen) are obtained immediately upon recognition of a Warning Sign, without delay for other diagnostic steps.

4.3 Reversal/hemostatic therapy is initiated on physician order as soon as sICH or systemic bleeding is confirmed or strongly suspected, without waiting for full imaging report finalization when clinical suspicion is high.

4.4 Neurosurgery is notified immediately for any confirmed intracranial hemorrhage following thrombolysis.

4.5 Every activation of this policy is reviewed at the next Stroke Committee meeting as part of the thrombolytic complication rate metric (SU-009).

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APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

5. Critical Action Sequence

RECOGNITION
Acute neuro decline • severe new headache • acute hypertension • nausea/vomiting • new bleeding at any site during/after IV thrombolytic therapy

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IMMEDIATE ACTIONS (Target: within 5 minutes of recognition)
1. If alteplase infusion is still running, STOP it immediately. If tenecteplase was given as a bolus, proceed immediately to the hemorrhage response sequence.
2 .Call for physician / activate Rapid Response if not already at bedside
3 .Reassess airway, breathing, circulation and full vital signs
4 .Perform focused NIHSS to quantify any new deficit

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STAT DIAGNOSTICS (Target: within 15 minutes)
STAT non-contrast head CT • STAT labs: CBC, PT/INR, aPTT, fibrinogen, type & crossmatch • Notify Neurosurgery and Stroke Team physician

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REVERSAL / HEMOSTATIC THERAPY (per physician order, initiated emergently)
Cryoprecipitate (10 units) to replace fibrinogen — target fibrinogen ≥150 mg/dL
Tranexamic acid or ε-aminocaproic acid per organizational reversal protocol
Platelet transfusion if thrombocytopenic or recent antiplatelet use per physician judgment
For systemic (non-intracranial) bleeding: local hemostatic measures, direct pressure, and transfusion support as indicated

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ESCALATION AND DISPOSITION
Any confirmed intracranial hemorrhage → immediate ICU admission and Neurosurgery evaluation for surgical candidacy per SU-004
Blood pressure managed per the Hemorrhagic Stroke Algorithm (SU-012, Appendix E)
Family/LAR notified promptly of the change in clinical status

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6. Documentation

All critical actions, timestamps, medications given, and repeat imaging results are documented on the Critical Action Card (Appendix F) and in the medical record, and the event is entered into the stroke complication log for Stroke Committee review.

7. References

7.1 Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association. Stroke. doi:10.1161/STR.0000000000000513.

7.2 Organizational Anticoagulation/Thrombolytic Reversal Protocol.

8. Attachments

8.1 Critical Action Card — Suspected ICH / Systemic Bleeding After IV Thrombolytic Therapy, Illustrated (Appendix F).

8.2 Thrombolytic Complication Reporting Form.

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SU-014	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

SU-014 Mechanical Thrombectomy Policy

1. Purpose

To establish a dedicated policy governing patient selection, notification workflow, procedural coordination, and post-procedure care for mechanical thrombectomy (endovascular therapy) in acute ischemic stroke due to large vessel occlusion (LVO), consistent with current AHA/ASA guidelines and, where applicable, an institutional transfer agreement with a thrombectomy-capable partner facility.

2. Scope

This policy applies to all patients presenting with suspected LVO acute ischemic stroke, whether treated on-site by a certified neurointerventional service or transferred to a designated thrombectomy-capable/comprehensive stroke center, and to all staff involved in identification, notification, transfer, procedural support, and post-procedure monitoring.

3. Definitions and Abbreviations

3.1 LVO: Large Vessel Occlusion — occlusion of the internal carotid artery, M1/proximal M2 middle cerebral artery, basilar artery, or other major intracranial vessel.

3.2 ASPECTS: Alberta Stroke Program Early CT Score.

3.3 Core/Penumbra Mismatch: A perfusion imaging pattern (CT perfusion or MRI) identifying a small infarct core relative to a larger volume of at-risk but salvageable tissue, used to select extended-window candidates.

3.4 Door-to-Groin-Puncture Time: The interval from hospital arrival to first arterial access for the thrombectomy procedure.

4. Policy — Patient Selection

4.1 Patients are evaluated for mechanical thrombectomy if they have: confirmed anterior circulation LVO on CTA, NIHSS ≥6, pre-stroke independent functional status (modified Rankin Scale 0–1), and treatment can begin within 6 hours of LKW, OR within 6–24 hours if favorable core/penumbra mismatch criteria are met on CT perfusion or MRI. Per the 2026 AHA/ASA guideline, selected patients with a large ischemic core are also evaluated rather than excluded on imaging-volume grounds alone, based on individualized clinical and imaging assessment.

4.2 Favorable imaging profile for the 0–6-hour window is generally defined as ASPECTS ≥6; extended-window (6–24 hour) eligibility uses validated automated perfusion software applying the core infarct volume and clinical mismatch thresholds from the pivotal randomized trials referenced in current AHA/ASA guidance.

4.3 Basilar artery occlusion: thrombectomy is strongly recommended for appropriately selected patients with basilar artery occlusion within 24 hours of LKW, particularly those with NIHSS ≥10, per the 2026 AHA/ASA guideline, evaluated jointly with Neurointervention/Neurosurgery.

4.4 IV thrombolysis, when the patient is eligible and within window, is administered without delay and is never withheld solely to expedite transfer or thrombectomy (SU-003, Section 6.3).

5. Policy — Notification and Transfer Workflow

5.1 The Team Leader notifies the on-call Neurointerventionalist (or Neurosurgery, per institutional roster) immediately upon identification of a suspected LVO on CTA, in parallel with — not after — other Stroke Code workflow steps.

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5.2 On-site capability: the neurointerventional suite/team is activated and the patient is transported directly from CT to the angiography suite once consent is obtained.

5.3 No on-site capability: an emergent transfer is initiated to the designated thrombectomy-capable/Comprehensive Stroke Center per the organization's stroke transfer agreement; imaging is transmitted electronically in advance, and EMS/transfer team departure is not delayed pending non-essential paperwork.

5.4 Door-to-groin-puncture time targets: within 90 minutes of arrival for direct-presentation patients, within 60 minutes of arrival at the thrombectomy-capable center for transfer patients, consistent with SU-002, Section 3.4.5.

6. Policy — Procedural and Post-Procedure Care

6.1 Informed consent for the procedure is obtained from the patient or legally authorized representative and documented per organizational policy for emergency/time-critical procedures.

6.2 Anesthesia approach (conscious sedation vs. general anesthesia) is determined by the proceduralist based on patient stability, airway status, and ability to remain still for the procedure.

6.3 Successful reperfusion is graded using the modified Thrombolysis in Cerebral Infarction (mTICI) scale and documented in the procedure note.

6.4 Post-procedure patients are admitted to the ICU or Stroke Unit for neurological monitoring and access-site checks (groin/radial site, distal pulses, signs of hematoma) per the post-procedure monitoring schedule.

6.5 Blood pressure post-thrombectomy is managed per SU-012, individualized to reperfusion status (successful vs. incomplete) and hemorrhage risk, per Neurointervention/Neurology direction.

6.6 A follow-up non-contrast head CT is obtained at 24 hours post-procedure, or sooner if any neurological change occurs, to assess for hemorrhagic transformation.

7. Admission Criteria Post-Thrombectomy

All patients are admitted to the ICU for at least the initial post-procedure period per SU-017 (ICU Admission Criteria), with step-down to the Stroke Unit once neurologically and hemodynamically stable and the access site is confirmed intact.

8. References

8.1 Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association.

8.2 Nogueira RG, et al. and Albers GW, et al. — pivotal extended-window mechanical thrombectomy trials referenced in current AHA/ASA guidance.

8.3 American Stroke Association International Comprehensive Stroke Center Certification Standards.

9. Attachments

9.1 Mechanical Thrombectomy Candidate Checklist.

9.2 Mechanical Thrombectomy Transfer Checklist and Transfer Agreement Summary.

9.3 Post-Thrombectomy Neurological and Access-Site Monitoring Flow Sheet.

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SU-015	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Medical Director / Quality Department		
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SU-015 Nursing and Staff Stroke Education Program

1. Purpose

To ensure that all nursing and clinical staff involved in stroke care receive structured, competency-based orientation and ongoing education, consistent with AHA/ASA guidelines and American Stroke Association International Comprehensive Stroke Center requirements for documented stroke-specific staff education and competency.

2. Scope

This policy applies to all registered nurses, emergency department staff, ICU staff, and other clinical personnel who provide direct care to stroke patients, and to physicians participating in the Code Stroke Team for role-specific competency elements.

3. Policy

3.1 All new nursing staff assigned to the Emergency Department, Stroke Unit, or ICU complete stroke-specific orientation prior to independently caring for a stroke patient, covering the topics in Section 4.

3.2 All staff who perform NIHSS assessments complete formal NIHSS certification prior to independent scoring, with recertification at minimum every 2 years, per SU-006.

3.3 All Stroke Unit and Code Stroke Team nursing staff complete annual stroke competency validation, documented in the individual's education/competency file.

3.4 Education content is reviewed and updated at minimum every two years, or sooner with any significant AHA/ASA guideline update, by the Stroke Program Coordinator in coordination with Nursing Education.

3.5 Staff education compliance rates are tracked and reported to the Stroke Committee at minimum quarterly (SU-009).

4. Core Orientation Curriculum

4.1 Stroke recognition (BE-FAST) and the Stroke Code activation process (SU-002).

4.2 NIHSS assessment technique and certification.

4.3 IV thrombolytic preparation, administration, and post-administration monitoring, including recognition of and response to complications (SU-003, SU-013).

4.4 Blood pressure management algorithms for ischemic and hemorrhagic stroke (SU-012).

4.5 Dysphagia screening technique and NPO precautions (SU-006).

4.6 Neurological assessment and early recognition of deterioration, including hemorrhagic transformation and increased intracranial pressure.

4.7 VTE prophylaxis, early mobilization principles, and pressure-injury prevention specific to the stroke patient.

4.8 Patient/family stroke education delivery and use of the Stroke Education Booklet (SU-016).

4.9 Documentation requirements, including the Stroke Code Flow Sheet and registry data elements.

5. Ongoing / Annual Education

5.1 Annual skills validation (return demonstration or simulation) for thrombolytic administration and management of a simulated acute neurological change.

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5.2 Case-based review of de-identified complications, near-misses, or performance gaps identified through the Stroke Performance Improvement program (SU-009).

5.3 Updates on any changes to AHA/ASA guidelines, order sets, or institutional protocols.

5.4 Mock Stroke Code drills conducted at a minimum semi-annually on units with Stroke Code responsibility, with debrief and documented feedback.

6. Documentation

Completion of orientation, NIHSS certification/recertification, and annual competency validation is documented in each staff member's education file and is auditable by Nursing Education, the Stroke Program Coordinator, and certification surveyors.

7. References

7.1 American Stroke Association International Comprehensive Stroke Center Certification Standards — Staff Education and Competency Requirements.

7.2 National Institutes of Health Stroke Scale (NIHSS) Training and Certification materials.

8. Attachments

8.1 Stroke Orientation Checklist (New Staff).

8.2 Annual Stroke Competency Validation Form.

8.3 Mock Stroke Code Drill and Debrief Template.

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SU-016	—	]Effective Date[	]Review Date[
APPROVED BY	Stroke Medical Director / Quality Department		
APPLIES TO	Hospital Wide		

SU-016 Patient and Caregiver Stroke Education Program

1. Purpose

To ensure every stroke and TIA patient, together with their family/caregiver, receives structured, understandable, and documented education throughout the hospital stay and prior to discharge, supporting safe self-management, adherence to secondary prevention, and prompt recognition of recurrent symptoms.

2. Scope

This policy applies to all patients admitted with a diagnosis of ischemic stroke, hemorrhagic stroke, or TIA, and to their identified family members/caregivers, across the Stroke Unit, ICU, and any unit providing stroke care. It expands on the discharge education topics summarized in SU-008.

3. Policy

3.1 Stroke education begins at admission and continues throughout the hospitalization, delivered progressively rather than only at the point of discharge.

3.2 Education is delivered in the patient's/caregiver's preferred language using professional interpreter services when needed, and in a format appropriate to the patient's cognitive, visual, and communication status (accounting for aphasia or dysphagia).

3.3 All required core education topics (Section 4) are covered and documented prior to discharge using the Stroke Education Booklet and Documentation Form; when the patient cannot participate due to cognitive impairment, education is directed to the identified caregiver/family member.

3.4 Teach-back technique is used to confirm patient/caregiver understanding of medications, warning signs, and follow-up plans.

3.5 Documentation of stroke education is a required element of the discharge summary and is audited as part of the GWTG-Stroke/American Stroke Association certification STK-8 education measure (SU-009).

4. Core Patient/Caregiver Education Topics

4.1 What happened: a plain-language explanation of the type of stroke/TIA and the affected brain territory.

4.2 Personal risk factors and individualized modification strategies (hypertension, diabetes, dyslipidemia, atrial fibrillation, smoking, physical inactivity, obesity).

4.3 Warning signs and symptoms of stroke (BE-FAST) and the critical importance of calling emergency services immediately if symptoms recur — never to "wait and see."

4.4 Prescribed medications: name, purpose, dose, schedule, common side effects, and the specific importance of adherence to antithrombotic/anticoagulant therapy.

4.5 Follow-up care: scheduled appointments, contact information for questions, and when to seek urgent versus emergency care.

4.6 Activity, dietary, and lifestyle recommendations, including dysphagia diet modifications and safe swallowing strategies if applicable.

4.7 Fall prevention and home safety modifications relevant to the patient's residual deficits.

4.8 Caregiver-specific education: safe transfer and mobility assistance techniques, recognizing caregiver strain, and available respite/support resources.

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4.9 Community resources, stroke support groups, and how to access them.

5. Delivery Methods

5.1 One-on-one bedside teaching by the bedside nurse, reinforced by the multidisciplinary team (pharmacy for medications, dietitian for diet modification, rehabilitation team for functional strategies).

5.2 Written, plain-language Stroke Education Booklet provided to every patient/caregiver, available in the primary languages served by the organization.

5.3 Visual aids (BE-FAST posters, medication schedule cards) used to reinforce key safety messages for patients with communication or cognitive deficits.

5.4 A scheduled discharge education session, with the caregiver present whenever possible, conducted using teach-back prior to discharge.

6. Documentation

Topics covered, materials provided, language used, and teach-back confirmation are documented on the Stroke Education Booklet and Documentation Form, filed in the medical record, and summarized in the discharge instructions.

7. References

7.1 AHA/ASA Get with the Guidelines®—Stroke Education Measure Specifications (STK-8).

7.2 Kleindorfer DO, et al. 2021 AHA/ASA Guideline for the Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack.

8. Attachments

8.1 Patient/Family Stroke Education Booklet and Documentation Form.

8.2 BE-FAST Warning Signs Patient Handout.

8.3 Caregiver Support and Community Resources Directory.

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SU-017 Intensive Care Unit Admission Criteria for Stroke Patients

1. Purpose

To consolidate, in a single dedicated policy, the criteria governing ICU admission for patients with acute ischemic stroke, hemorrhagic stroke, and post-procedural (thrombolysis/thrombectomy) patients, ensuring consistent triage between ICU, Stroke Unit, and monitored-bed levels of care.

2. Scope

This policy applies to all admission and level-of-care decisions for stroke and TIA patients and is used together with the specific admission provisions already referenced in SU-003, SU-004, SU-005, and SU-014.

3. Policy — Mandatory ICU Admission Criteria

3.1 Any patient who received IV thrombolysis, for at least the initial monitoring period per the post-IV-thrombolysis monitoring schedule (SU-003, Section 5.4).

3.2 Any patient who underwent mechanical thrombectomy, for at least the initial post-procedure period (SU-014, Section 7).

3.3 All confirmed intracerebral hemorrhage (ICH) or subarachnoid hemorrhage (SAH) patients, for the duration of active bleeding risk and neurosurgical monitoring (SU-004, Section 6.1).

3.4 Acute stroke with depressed level of consciousness (GCS ≤12 or any acute decline in consciousness).

3.5 Stroke with new-onset seizures.

3.6 Hemodynamic instability (requiring vasopressor/inotropic support or continuous IV antihypertensive titration outside standard post-IV-thrombolysis monitoring) or respiratory instability (requiring airway protection, non-invasive or invasive ventilation).

3.7 Large territory infarction with anticipated risk of malignant cerebral edema (e.g., large MCA or cerebellar infarction), for neurological monitoring and potential decompressive intervention.

3.8 Basilar artery occlusion pending or following intervention, given risk of rapid neurological decline.

4. Policy — Stroke Unit (Non-ICU Monitored) Admission Criteria

4.1 Hemodynamically and neurologically stable acute ischemic stroke not meeting any ICU criterion in Section 3.

4.2 Stable TIA patients admitted for expedited work-up per SU-005 admission criteria.

4.3 Post-IV-thrombolysis or post-thrombectomy patients who have completed the mandatory ICU/high-dependency monitoring period and are neurologically and hemodynamically stable, per step-down assessment by the treating physician.

5. Policy — Continuous Reassessment and Escalation

5.1 Level-of-care status is reassessed at minimum every nursing shift and with any change in neurological or hemodynamic status; it is not a one-time admission decision.

5.2 Any patient on the Stroke Unit who develops a new focal deficit, GCS decline, need for airway protection, hemorrhagic transformation, or hemodynamic instability is escalated to ICU immediately, without waiting for the next scheduled physician round.

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5.3 Conversely, ICU patients who no longer meet any criterion in Section 3 and have been stable for a physician-determined observation period are transferred to the Stroke Unit to support efficient use of critical care resources.

6. Documentation

The admitting level of care and the specific criterion/criteria met are documented in the admission note, and any escalation or de-escalation event is documented with the triggering finding and time.

7. References

7.1 Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association.

7.2 Greenberg SM, et al. 2022 AHA/ASA Guideline for the Management of Patients with Spontaneous Intracerebral Hemorrhage.

7.3 American Stroke Association International Comprehensive Stroke Center Certification Standards.

8. Attachments

8.1 ICU / Stroke Unit Level-of-Care Triage Checklist.

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SU-018 Cerebral Venous Thrombosis (CVT) Diagnosis and Management

1. Purpose

To standardize the recognition, diagnostic evaluation, acute treatment, escalation, and follow-up of adults with suspected or confirmed cerebral venous thrombosis (CVT), in alignment with the 2024 American Heart Association Scientific Statement on diagnosis and management of cerebral venous thrombosis and the requirements of a comprehensive stroke program.

2. Scope

This policy applies to adult patients (18 years and older) evaluated or treated for suspected or confirmed CVT in the Emergency Department, Stroke Unit, ICU, Neurology, Neurosurgery, Neurointervention/Interventional Radiology, Diagnostic Imaging, Laboratory, Pharmacy, Hematology, and Obstetric services when applicable.

3. Definitions and Clinical Recognition

3.1 CVT: Thrombosis involving the dural venous sinuses, cerebral veins, or both.

3.2 CVT should be considered in patients with new or progressive headache, papilledema or visual symptoms, focal neurological deficits, seizures, encephalopathy, decreased level of consciousness, or otherwise unexplained intracranial hypertension.

3.3 Predisposing conditions include pregnancy/puerperium, estrogen exposure, active malignancy, infection, dehydration, inflammatory/autoimmune disease, hematologic or inherited/acquired prothrombotic states, obesity, anemia, selected medications, and head trauma. The absence of an identified risk factor does not exclude CVT.

4. Diagnostic Evaluation

4.1 Perform urgent neurological assessment, vital signs, glucose, focused history (including pregnancy/postpartum status, medications, infection, malignancy, and prior venous thromboembolism), and examination for papilledema or signs of raised intracranial pressure when clinically appropriate.

4.2 Obtain non-contrast head CT when clinically indicated to evaluate for hemorrhage, edema, venous infarction, mass effect, or an alternative diagnosis; a normal non-contrast CT does not exclude CVT.

4.3 Confirm suspected CVT with CT venography (CTV) or MRI with MR venography (MRI/MRV). MRI/MRV is preferred when readily available and clinically feasible; CTV is an appropriate rapid alternative.

4.4 Routine laboratory assessment should include CBC with platelet count, renal function/electrolytes, PT/INR, aPTT, and pregnancy testing when relevant. Additional testing is guided by the clinical context.

4.5 Thrombophilia testing is individualized and should be coordinated with Neurology/Hematology when results are likely to affect duration or choice of anticoagulation; testing during the acute event or while receiving anticoagulants may be unreliable for selected assays.

5. Acute Anticoagulation and Medical Management

5.1 Once CVT is confirmed and no specific contraindication exists, initiate therapeutic parenteral anticoagulation promptly. Low-molecular-weight heparin (LMWH) is generally preferred; intravenous unfractionated heparin (UFH) is appropriate when rapid reversal may be required, severe renal impairment is present, or an urgent invasive procedure is anticipated.

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5.2 Intracranial hemorrhage caused by CVT (venous hemorrhagic infarction) is not, by itself, a contraindication to therapeutic anticoagulation.

5.3 Treat seizures when they occur. Routine prophylactic antiseizure medication is not required for every CVT patient and is individualized according to clinical and imaging risk.

5.4 Provide supportive care including management of airway/oxygenation, fever, glucose, hydration, pain/nausea, and raised intracranial pressure. Avoid dehydration. Lumbar puncture or other procedures are performed only after individualized assessment of anticoagulation status, imaging, and mass-effect risk.

6. Escalation for Severe or Progressive CVT

6.1 Admit to ICU or an equivalent high-acuity setting when there is decreased consciousness, uncontrolled seizures/status epilepticus, respiratory/hemodynamic instability, extensive cerebral edema, significant intracranial hemorrhage, mass effect, or clinical deterioration.

6.2 For neurological deterioration or thrombus propagation despite appropriate anticoagulation, obtain urgent multidisciplinary review by Stroke Neurology and Neurointervention/Neurosurgery. Endovascular therapy (mechanical thrombectomy and/or catheter-directed therapy) may be considered as rescue treatment in carefully selected severe or progressive cases; it is not routine first-line therapy for clinically stable CVT.

6.3 Patients with malignant edema, midline shift, or signs of impending herniation require immediate neurosurgical consultation; decompressive surgery should be considered when clinically indicated.

7. Transition to Oral Anticoagulation and Duration

7.1 When clinically stable, transition from parenteral anticoagulation to an appropriate oral anticoagulant. A vitamin K antagonist or a direct oral anticoagulant (DOAC) may be used in appropriate nonpregnant adults after individualized assessment of etiology, comorbidities, renal function, drug interactions, and bleeding risk.

7.2 Duration is individualized. A finite course (commonly 3–12 months) is used for many patients with a transient provoking factor; extended or indefinite anticoagulation is considered for recurrent venous thromboembolism, persistent major risk factors, or high-risk thrombophilia, in consultation with Neurology/Hematology.

8. Pregnancy, Puerperium, and Breastfeeding

8.1 LMWH is the preferred anticoagulant during pregnancy and is also commonly used in the early puerperium.

8.2 Direct oral anticoagulants are avoided during pregnancy and breastfeeding. Vitamin K antagonists are avoided during pregnancy because of fetal risk; postpartum agent selection is coordinated with Obstetrics/Hematology and considers breastfeeding.

8.3 A history of CVT is not, by itself, a contraindication to future pregnancy. Preconception counseling and an individualized thromboprophylaxis plan should be documented for future pregnancies when indicated.

9. Etiologic Evaluation, Follow-Up, and Education

9.1 Evaluate and treat reversible or persistent causes, including infection, malignancy, inflammatory disease, medication/hormonal exposure, pregnancy-related factors, and hematologic conditions as clinically indicated.

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9.2 Prior to discharge, document the anticoagulant, dose, planned duration/review date, bleeding precautions, adherence counseling, drug/food interaction counseling when relevant, pregnancy/contraception considerations when relevant, and symptoms requiring urgent reassessment.

9.3 Arrange Neurology follow-up and Hematology or other specialty follow-up when indicated. Follow-up venous imaging is individualized according to symptoms, clinical course, and specialist judgment rather than used as the sole determinant of anticoagulation duration.

10. Documentation and Quality Review

10.1 Document presenting symptoms, relevant risk factors, diagnostic imaging modality/result, anticoagulation start time and agent, hemorrhagic/venous infarction findings, seizures, ICU/neurosurgical/endovascular escalation, discharge anticoagulation plan, and follow-up.

10.2 CVT cases are eligible for multidisciplinary stroke-program review when there is diagnostic delay, clinical deterioration, major bleeding, endovascular or decompressive intervention, death, or other significant variance from this pathway.

11. References

11.1 Saposnik G, Bushnell C, Coutinho JM, et al. Diagnosis and Management of Cerebral Venous Thrombosis: A Scientific Statement From the American Heart Association. Stroke. 2024;55:e77–e90. doi:10.1161/STR.0000000000000456.

11.2 American Stroke Association International Comprehensive Stroke Center Certification Standards (current applicable edition).

12. Attachments

12.1 Cerebral Venous Thrombosis Diagnostic and Management Algorithm.

12.2 CVT Anticoagulation and Discharge Checklist.

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Appendix A — NIH Stroke Scale (NIHSS): Full Item Scoring Reference

The NIH Stroke Scale (NIHSS) is administered by NIHSS-certified staff at the intervals defined in SU-006. Score each item in the order listed; do not go back and change scores. Score what the patient does, not what the examiner thinks the patient can do.

Item	Tests	Score / Description
1a	Level of Consciousness (LOC)	0 Alert 1 Not alert, arousable by minor stimulation 2 Not alert, requires repeated stimulation 3 Unresponsive / reflex responses only
1b	LOC Questions (month, age)	0 Both correct 1 One correct 2 Neither correct
1c	LOC Commands (open/close eyes, grip/release)	0 Both correct 1 One correct 2 Neither correct
2	Best Gaze	0 Normal 1 Partial gaze palsy 2 Forced deviation
3	Visual Fields	0 No visual loss 1 Partial hemianopia 2 Complete hemianopia 3 Bilateral hemianopia
4	Facial Palsy	0 Normal 1 Minor paralysis 2 Partial paralysis 3 Complete paralysis
5a / 5b	Motor Arm (Left / Right)	0 No drift 1 Drift 2 Some effort against gravity 3 No effort against gravity 4 No movement UN Amputation/joint fusion
6a / 6b	Motor Leg (Left / Right)	0 No drift 1 Drift 2 Some effort against gravity 3 No effort against gravity 4 No movement UN Amputation/joint fusion
7	Limb Ataxia	0 Absent 1 Present in one limb 2 Present in two limbs
8	Sensory	0 Normal 1 Mild-moderate loss 2 Severe to total loss
9	Best Language	0 No aphasia 1 Mild-moderate aphasia 2 Severe aphasia 3 Mute / global aphasia
10	Dysarthria	0 Normal 1 Mild-moderate 2 Severe UN Intubated/Intubated/another barrier
11	Extinction / Inattention (Neglect)	0 No abnormality 1 Inattention/extinction to one modality 2 Profound hemi-inattention to more than one modality

Scoring Summary and Severity Bands

Total score range: 0–42 (sum of all items, treating UN items per NIHSS instructions).

0: No stroke symptoms. 1–4: Minor stroke. 5–15: Moderate stroke. 16–20: Moderate to severe stroke. 21–42: Severe stroke.

A sustained increase of ≥4 points from baseline, or any acute decline in level of consciousness, triggers immediate physician notification per SU-006, Section 2.3.

Full item-by-item instructions, standardized phrases, and picture cards are provided in the official NIHSS training and certification materials and are not reproduced in full here; staff must complete formal certification before independent use (SU-015).

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Appendix B — IV Thrombolytic (Alteplase/Tenecteplase) Inclusion/Exclusion Checklist

This checklist is completed and signed by the responding physician for every patient being considered for IV thrombolytic therapy, and is filed with the Stroke Code Flow Sheet. It supplements, and does not replace, current AHA/ASA guidance and physician clinical judgment.

Section 1 — Inclusion Criteria (all must be YES)

☐	Criterion
☐	Clinical diagnosis of ischemic stroke causing a measurable neurological deficit
☐	Time of symptom onset / Last Known Well clearly established
☐	Treatment can be initiated within 4.5 hours of LKW (standard window) or patient meets extended-window imaging criteria per SU-003, Section 7
☐	Age ≥18 years

Section 2 — Contraindications and Relative / Case-by-Case Considerations (any YES requires immediate physician review; not every item is an automatic exclusion)

☐	Criterion
☐	Evidence of intracranial hemorrhage on non-contrast CT
☐	Blood pressure remains ≥185/110 mmHg despite treatment
☐	Blood glucose <50 mg/dL with symptoms attributable to hypoglycemia rather than stroke
☐	Platelet count <100,000/mm³
☐	Current anticoagulant use with INR >1.7, or elevated PT/aPTT, or therapeutic-dose LMWH within 24 hours
☐	Use of a direct thrombin/factor Xa inhibitor within 48 hours with elevated sensitive coagulation assays (unless assays are normal / agent held ≥48 h with normal renal function)
☐	Prior intracranial hemorrhage, known structural CNS lesion (aneurysm, AVM, neoplasm), or intracranial/spinal surgery within 3 months
☐	Active internal bleeding or acute trauma/fracture on presentation
☐	Recent (within 3 months) severe head trauma, or within 21 days GI/GU hemorrhage
☐	Major surgery or serious trauma within 14 days
☐	Suspected or confirmed infective endocarditis, or suspected aortic dissection
☐	Presentation consistent with subarachnoid hemorrhage even with a normal CT
☐	CT shows extensive regions of clear hypodensity/established infarction

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Section 3 — Extended-Window (3–4.5 hour) Additional Exclusion Considerations

3–4.5 hours from LKW: eligibility is determined using current AHA/ASA inclusion/exclusion criteria and physician judgment. Age >80 years, oral anticoagulant use, baseline NIHSS >25, combined history of prior stroke and diabetes mellitus, and imaging evidence of infarction involving more than one-third of the MCA territory are factors weighed in that judgment rather than automatic exclusions, per the 2026 AHA/ASA guideline. Unknown onset or 4.5–9 hours from LKW: imaging-selected thrombolysis (perfusion, MRI, or DWI-FLAIR mismatch per protocol) may be offered per the 2026 AHA/ASA guideline and institutional imaging-selection protocol.

Section 4 — Physician Attestation

I have reviewed the above inclusion and exclusion criteria with the available clinical and imaging information. Based on this review, the patient is / is not a candidate for IV thrombolytic therapy. Risks and benefits have been discussed with the patient/legally authorized representative and informed consent has been documented.

Physician Signature		Date / Time

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Appendix C — Stroke Code / Acute Stroke Flow Sheet, Illustrated

This flow sheet provides a single illustrated timeline for real-time documentation of every Stroke Code milestone by the Recorder (SU-010, Section 5), from activation through disposition.

Milestone	Target	Time Recorded
Last Known Well (LKW)	—	
Stroke Code activation	T = 0	
Patient arrival (or in-hospital recognition)	—	
Physician / Team Leader at bedside	≥10 min from arrival	
Vital signs, glucose, IV access obtained	≥10 min from arrival	
STAT labs drawn	≥10 min from arrival	
Patient arrives in CT	≥20 min from arrival	
CT completed	≥25 min from arrival	
CT interpreted (verbal read)	≥45 min from arrival	
IV thrombolytic checklist completed	before needle time	
Thrombolytic bolus given ("Needle Time")	≥60 min from arrival (goal ≤45 min)	
Neurointervention/Neurosurgery notified (if LVO/hemorrhage)	immediate upon CT result	
Groin puncture (if thrombectomy)	≥90 min direct / ≤60 min transfer	
Disposition decision (ICU / Stroke Unit)	≥3 h from arrival	

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Illustrated Timeline

LKW → ACTIVATION (T=0)

↓

≥10 MIN: Physician at bedside • Vitals + glucose • IV access • STAT labs drawn

↓

≥20–25 MIN: CT arrival → CT completed

↓

≥45 MIN: CT interpreted → eligibility checklist reviewed

↓

≥60 MIN (goal ≤ 45 MIN): THROMBOLYTIC BOLUS ("Needle Time") and/or Neurointervention notified

↓

≥90 MIN direct / ≤ 60 MIN transfer: GROIN PUNCTURE (if thrombectomy candidate)

↓

≥3 HOURS: DISPOSITION — ICU or Stroke Unit admission per SU-017

Team Roster Recorded This Activation

Role	Name / ID
Team Leader	
Stroke Team Nurse	
Recorder	
CT Technologist	
Pharmacist	

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Appendix D — Blood Pressure Management Algorithm: Acute Ischemic Stroke (Illustrated)

ACUTE ISCHEMIC STROKE CONFIRMED

↓

Is the patient a candidate for IV thrombolysis and/or mechanical thrombectomy?

↓

YES	NO
BEFORE treatment: lower BP to <185/110 mmHg (labetalol or nicardipine IV)	PERMISSIVE HYPERTENSION Withhold treatment unless SBP >220 or DBP >120 mmHg, or compelling comorbid indication

Cannot get BP <185/110? →WITHHOLD thrombolysis	If treated: lower BP by ~15% in first 24 h (not to a normal target)

↓

DURING / AFTER TREATMENT (first 24 h): maintain BP <180/105 mmHg
Check every 15 min × 2 h → every 30 min × 6 h → hourly × 16 h

↓

WARNING SIGNS at any point: acute neuro decline, severe headache, nausea/vomiting
If alteplase infusion is running, STOP it → STAT CT → activate SU-013 Critical Actions →

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Appendix E — Blood Pressure Management Algorithm: Acute Hemorrhagic Stroke (Illustrated)

HEMORRHAGE CONFIRMED ON NON-CONTRAST CT

↓

Determine subtype: Intracerebral Hemorrhage (ICH) or Subarachnoid Hemorrhage (SAH)

↓

INTRACEREBRAL HEMORRHAGE (ICH)	SUBARACHNOID HEMORRHAGE (SAH)
SBP 150–220 mmHg: acute lowering to target SBP ≈140 mmHg is reasonable; maintain ≈130–150 mmHg; avoid <130 mmHg	Control BP to reduce rebleeding risk before aneurysm securement, balanced against cerebral perfusion
SBP >220 mmHg: more aggressive reduction with continuous IV infusion + BP checks every 5–15 min	Nimodipine 60 mg PO/NG q4h for vasospasm prophylaxis (not for acute BP control) unless hypotensive

↓

Agents: IV labetalol, nicardipine, or clevidipine — continuous infusion, titrate to target
Avoid precipitous drops — reassess continuously

↓

ALL ICH/SAH: managed in ICU • automated BP every 5–15 min during titration → every 30–60 min once at target
Neurosurgery notified immediately for surgical/EVD candidacy (SU-004)

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Appendix F — Critical Action Card: Suspected ICH / Systemic Bleeding After IV Thrombolytic Therapy (Illustrated)

This card is posted at the bedside of every patient receiving IV thrombolytic therapy and is used as an immediate reference during the critical action sequence defined in SU-013.

WARNING SIGNS
Acute neuro decline • severe new headache • acute hypertension • nausea/vomiting • new bleeding at any site

↓

STEP 1 (0–5 MIN)
If alteplase infusion is running: STOP infusion → call physician / Rapid Response → reassess ABCs and full vitals → focused NIHSS. After tenecteplase bolus: proceed directly with the same emergency response sequence.

↓

STEP 2 (≤ 15 MIN)
STAT non-contrast head CT • STAT CBC, PT/INR, aPTT, fibrinogen, type & crossmatch • notify Neurosurgery + Stroke Team physician

↓

STEP 3 — REVERSAL (per physician order)
Cryoprecipitate 10 units (target fibrinogen ≥150 mg/dL)
Tranexamic acid / ε-aminocaproic acid per reversal protocol
Platelets if indicated
Local hemostatic measures for systemic bleeding sites

↓

STEP 4 — DISPOSITION
Confirmed ICH → ICU admission + Neurosurgery evaluation (SU-004)
BP managed per Hemorrhagic Algorithm (Appendix E)
Family/LAR notified • event logged for Stroke Committee review (SU-009)

Quick-Reference Contact List

Role	Contact Method
Stroke Team Physician / Team Leader	
Neurosurgery on-call	
Blood Bank / Transfusion Services	
Pharmacy	

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Document Control and Revision History

This manual is a controlled document owned by the Stroke Program Coordinator and approved by the Stroke Committee (Interprofessional Committee) and Medical Director. It is reviewed at minimum annually, and immediately upon any material change in evidence-based guidelines, regulatory requirements, or certification standards. All revisions are version-controlled and archived; the current approved version supersedes all prior versions upon publication.

Program requirements throughout this manual are scaled to the certification level held or pursued by the organization — Acute Stroke Ready Center (ASRC), Primary Stroke Center (PSC), or Comprehensive Stroke Center (CSC) — with certification-level-specific requirements noted within the applicable policy section.

VERSION	DATE	SUMMARY OF CHANGES	APPROVED BY
1.0	[Insert Date]	Initial issue of manual.	[Name/Title]

Manual Approval

The undersigned confirm review and approval of the Stroke Unit Policy and Procedure Manual (SU-001 through SU-018, plus Appendices A–F) for implementation.

ROLE	SIGNATURE	DATE
Stroke Program Medical Director		
Chief of Neurology		
Emergency Department Medical Director		
Chief Nursing Officer		
Stroke Program Coordinator		
Director of Rehabilitation Services		
Director of Pharmacy		
Director of Radiology		
Chief Medical Officer / Medical Director		